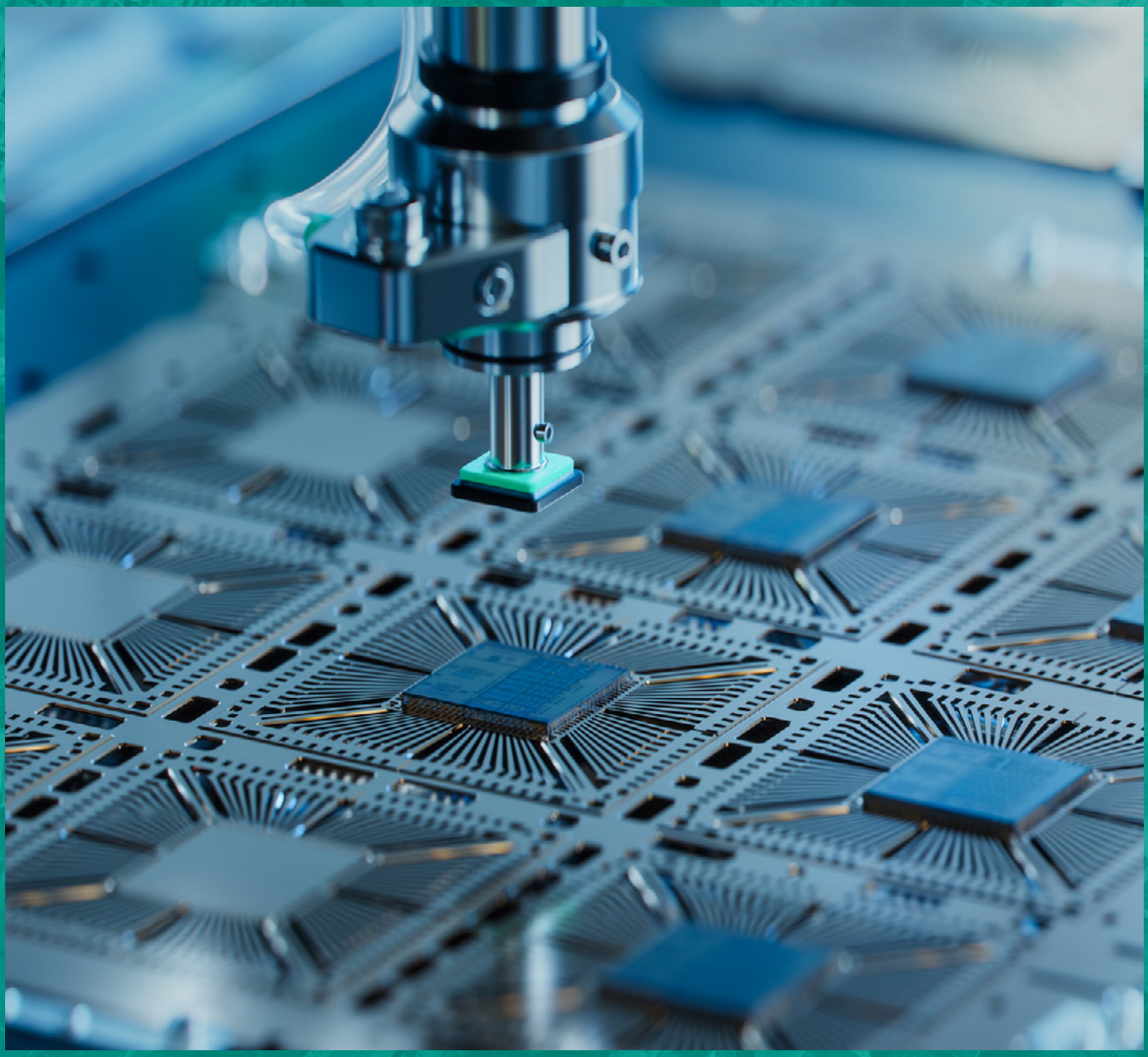


Additional sector guidance
Technology and communications

September 2026
Version 1.0

SICS® industry: Electronic manufacturing services & original design manufacturing (TC-ES); Hardware (TC-HW); Internet media & services (TC-IM) Semiconductors (TC-SC); Software & IT services (TC-SI); Telecommunication services (TC-TL)

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Taskforce on Nature-related Financial Disclosures

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1. Introduction

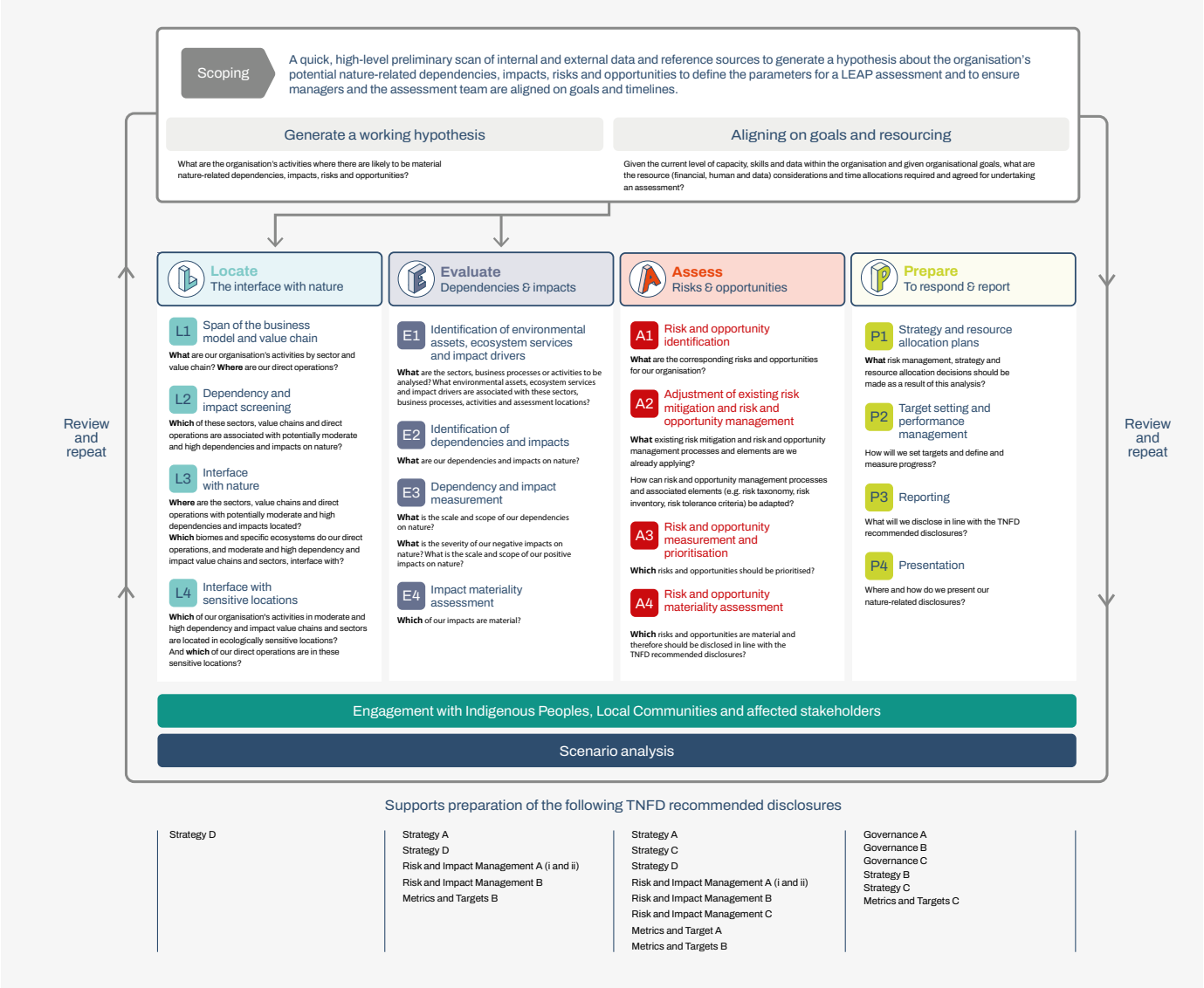
1.1. The purpose of this guidance

In September 2023, the TNFD published its recommendations for disclosure of nature-related issues and supporting implementation guidance. This document provides additional guidance for the technology and communications sector, covering:

- The **assessment of nature-related issues** (dependencies, impacts, risks and opportunities) using the TNFD's LEAP approach (Section 2); and
- The **disclosure of sector-specific metrics** in line with the TNFD's recommended approach to metrics (Section 3).

The TNFD's Guidance on the identification and assessment of nature-related issues: The LEAP approach is designed as an iterative process – across business locations and business lines – in line with established risk management processes and corporate reporting cycles. Organisations may choose to start with a narrow scope for a LEAP assessment and gradually expand the scope of the assessment as they gain experience and insight.

Figure 1: The TNFD approach for identification and assessment of nature-related issues – LEAP



The TNFD recognises that there can be significant differences across sectors for organisations applying the LEAP approach. It has published this additional guidance with input from knowledge partners and market participants to help organisations in the technology and communications sector apply the LEAP approach to their context.

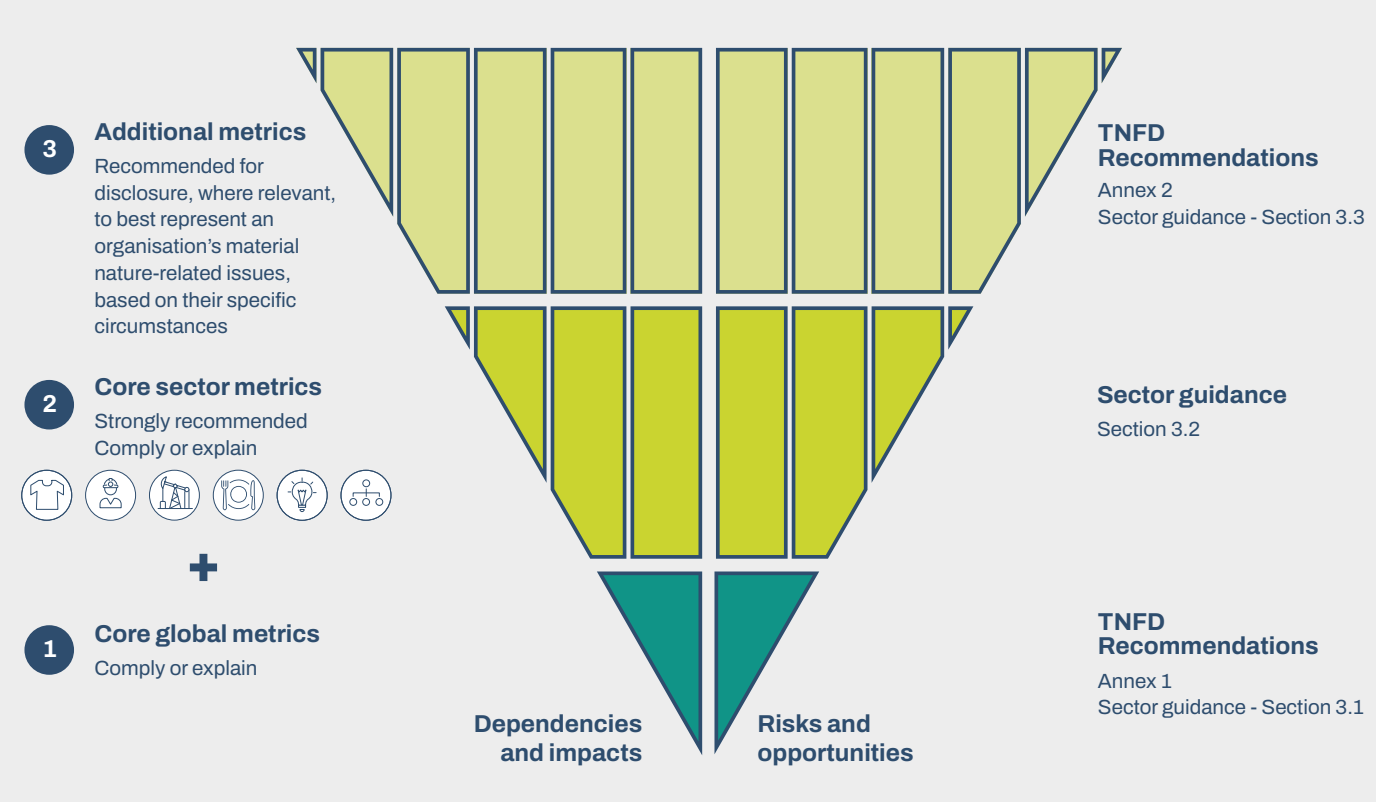
The structure of the LEAP approach is set out in Figure 1. The guidance in Section 2 follows that structure.

The Taskforce recognises that investors and other stakeholders require quantitative information to compare performance and nature-related issues within sectors. To facilitate that analysis, Section 3 covers:

- Guidance on the application of the core global disclosure indicators and metrics to the technology and communications sector (Section 3.1); and
- Core and additional sector disclosure indicators and metrics (Sections 3.2 and 3.3).

Figure 2 provides an overview of the TNFD disclosure measurement architecture and where indicators and metrics are listed in the TNFD recommendations and in the sector guidance.

Figure 2: TNFD disclosure metrics architecture signposted to metrics list



The guidance in Section 3 expands on the disclosure indicators and metrics outlined in Annexes 1 and 2 of the TNFD recommendations and adds sector-specific information where relevant. The TNFD has incorporated and sought to build on existing industry standards and disclosure metrics wherever possible to leverage current data collection and reporting practices and minimise assessment and reporting costs.

1.2. Audience for this guidance

This guidance covers organisations with business models or value chains in the Sustainable Industry Classification System (SICS®) Technology & Communications sector. It is relevant to all six industries in this sector: Electronic Manufacturing Services & Original Design Manufacturing (EMS & ODM); Hardware; Internet Media & Services; Semiconductors; Software & IT Services; and Telecommunication Services, as outlined in Annex 1.

Figure 3: Technology and communications value chain

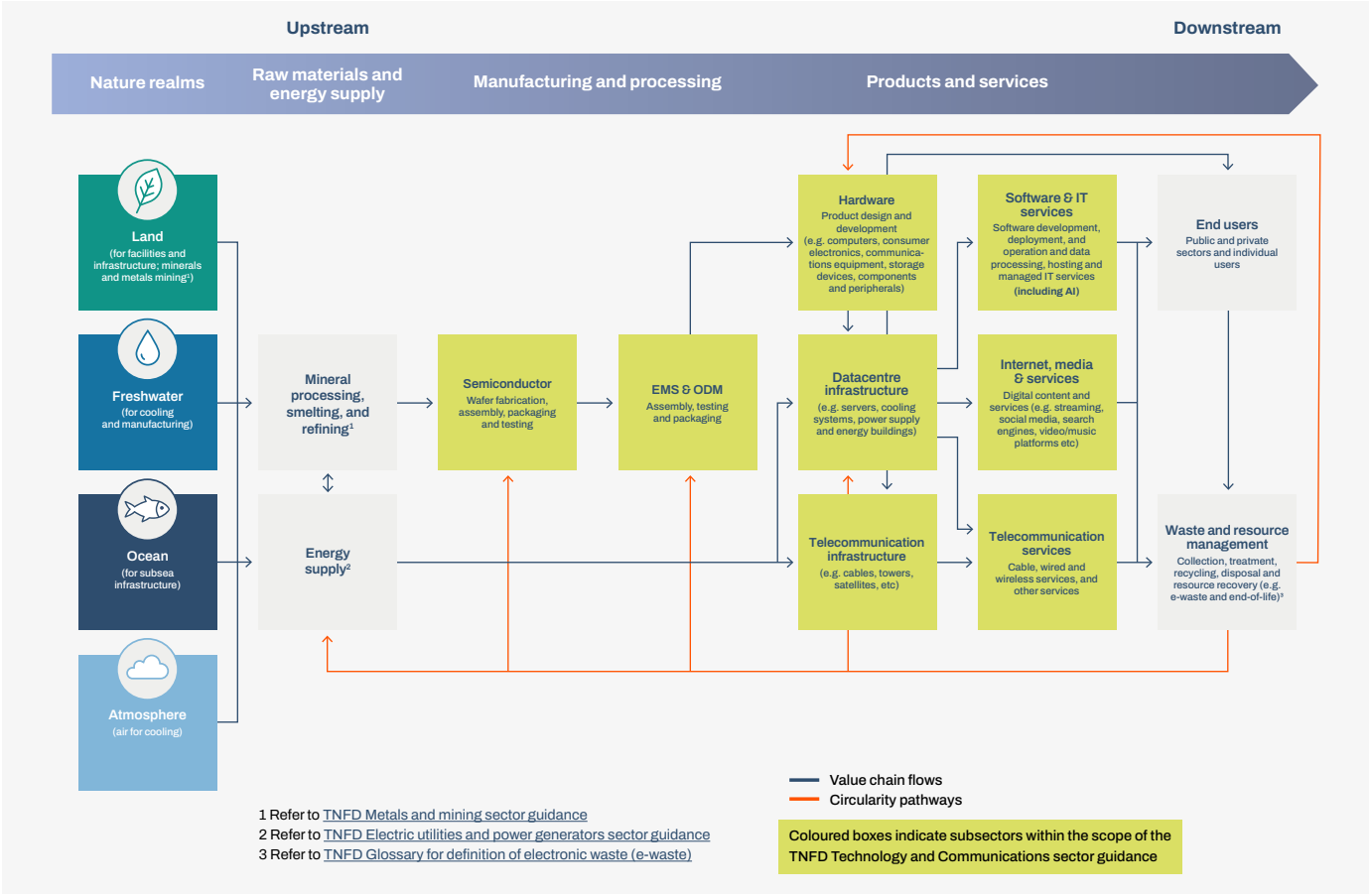


Figure 3 illustrates the value chain that organisations should consider as part of their value chain mapping from a nature perspective, highlighting where nature-related dependencies and impacts may arise. From left to right, the value chain shows raw materials and energy supply, manufacture and processing, products and services, users from public and private sectors and individual consumers, and finally resource and waste management.

In products and services, this value chain makes explicit the physical infrastructure of datacentres (formally within Software and IT services in SICS®) and telecommunications infrastructure (part of Telecommunication Services in SICS®). Artificial Intelligence (AI) models, systems and applications fall within Software & IT services in SICS® (see Box 1 for more on AI in the value chain).

The value chain in Figure 3 above reflects the predominant land-based configuration of datacentre infrastructure. Emerging deployment models include subsea datacentres and proposed space-based datacentres. Where relevant, organisations should reflect such deployment models and associated enabling infrastructure in their value chain mapping and nature-related assessments.

Box 1: Artificial Intelligence (AI) in the value chain: connecting software to physical infrastructure

Under SICS®, companies whose core business is developing and providing Artificial Intelligence (AI) models would generally be classified within Software & IT Services. The industry description includes cloud computing, big data analysis and machine learning, which together capture key elements of the software layer of AI. From a nature perspective, this guidance considers what enables AI: a physical value chain that depends on and impacts nature.

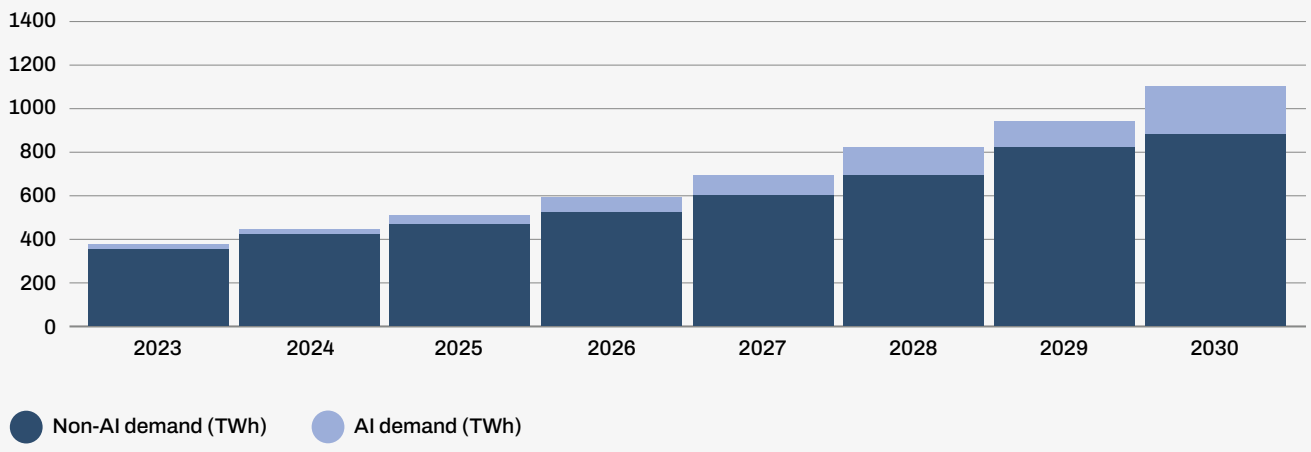
Developing and using AI requires substantial computing capacity, especially for training and running large language models at scale. This capacity is provided by datacentres, telecommunications infrastructure, semiconductors and other hardware. Across their construction, operation and end-of-life, these systems require land, energy, water, minerals and metals, as well as ecosystem services such as water purification and soil and sediment retention. These systems can generate impacts through drivers such as land use change, water use, pollution and waste.

Rapid adoption of AI (in the technology and communications sector, and in other sectors) is accelerating demand for this infrastructure. Figure 4 illustrates this growth in datacentre demand, with AI accounting for a growing share (see Section 1.3 for further context on the scale and growth of AI). This growth amplifies dependencies and impacts on nature across the value chain, which may give rise to nature-related risks for any organisation that develops or uses AI, regardless of sector. AI may also provide nature-related opportunities, including through improved monitoring, resource efficiency and decision making.

For a nature-related assessment, organisations developing or using AI should look beyond AI as software and consider the full physical value chain that enables it, including their dependencies and impacts on nature resulting from developing and using AI.

Sources: OECD (2026) AI use by individuals surges across the OECD as adoption by firms continues to expand, IEA (2026) Key Questions on Energy and AI and WEF (2026) Data Centre Sustainability and Resilience: A Decision Playbook for AI Infrastructure.

Figure 4: Growth in global data centre electricity demand (total and for AI)



Note: This World Economic Forum scenario extrapolates the IEA's 2023–2026 forecast to 2030, using 2021–2023 historical growth, the projected 2023–2026 growth rate and other sources.

Source: WEF (2025), Artificial Intelligence's Energy Paradox: Balancing Challenges and Opportunities, based on multiple sources including the International Energy Agency (IEA), Goldman Sachs and Accenture.

This guidance is a supplement to the TNFD's Guidance on the identification and assessment of nature-related issues: The LEAP approach and should be read in conjunction with that guidance. Organisations in the technology and communications sector may also find it helpful to refer to the TNFD biome guidance and Guidance on engagement with Indigenous Peoples, Local Communities and affected stakeholders to support their assessment. They should also consider interactions with other sectors and can refer to TNFD sector guidance on metals and mining, engineering, construction and real estate, construction materials and electric utilities and power generators, among others.

The examples in this guidance are intended to be illustrative. They are not exhaustive, universally applicable or recommended for all organisations within the sector. Each organisation's context, location and interactions with nature are unique. The TNFD encourages all organisations to consult additional sources, including scientific references, industry standards or best practice guides to conduct thorough assessments specific to their operations and value chains. This guidance aims to support, not replace, a tailored assessment, which will be necessary for each organisation.

Table 1: Areas of LEAP with additional guidance for the technology and communications sector in this guidance document

Scoping	✓

L1	✓	E1	✓	A1	✓	P1	✓
L2	✓	E2	✓	A2		P2	✓
L3	✓	E3	✓	A3		P3	✓
L4	✓	E4		A4		P4	

1.3. Sector background

Technology's growing role in the global economy

Technology and communications underpin almost every aspect of modern economic activity and daily life. Demand for digital infrastructure, electronic devices, data processing and connectivity has grown rapidly and is expected to continue to grow, driven by digitalisation, electrification and cloud computing. Artificial Intelligence (AI) further accelerates this growth (see Box 1 and Figure 4).[1]

Across technology industries, the scale and growth of economic activity are substantial:

- The global AI market is projected to surge from USD 189 billion in 2023 to USD 4.8 trillion by 2033, a 25-fold increase in just a decade;[2]
- More than 11,000 datacentres are operating worldwide, with demand for datacentre capacity expected to grow by approximately 20% per year to 2030,[3] and global datacentre build-out forecast to require around USD 3 trillion by 2030;[4]
- Around one trillion semiconductors are sold annually,[5] with recent production concentrated in a limited number of geographies,[6] and global semiconductor chip fabrication plant spending forecast to exceed USD 1.5 trillion from 2024 to 2030;[7] and
- The size of the electronic manufacturing services market stood at approximately USD 650 billion in 2025 and is forecast to expand to over USD 860 billion by 2030 at a nearly 6% compound annual growth rate.[8]

1 OECD (n.d.) Digital transformation. Policy Issue.

2 UNCTAD (2025) AI market projected to hit $4.8 trillion by 2033, emerging as dominant frontier technology.

3 Barclays Investment Bank (2025) AI revolution: Meeting massive AI infrastructure demands.

4 JLL (2026) 2026 Global Data Center Outlook.

5 SIA (2019) Global Semiconductor Sales Increase 13.7 Percent to $468.8 Billion in 2018.

6 OECD (2025) The chip landscape geographical distribution of wafer fabrication capacity.

7 PwC (2026) Semiconductor and beyond 2026.

8 Mordor Intelligence (2026) Electronics Manufacturing Services Market Size & Share Analysis – Growth Trends and Forecast (2026 – 2031).

The sector's value chain and nature

Nature-related dependencies and impacts may arise across the technology and communications value chain (see Figure 3), from raw material extraction and semiconductor manufacturing to hardware manufacture, datacentres, telecommunications infrastructure, software and IT services, internet media services and resource and waste management.

Dependencies and impacts on nature can occur outside an organisation's direct operational control, while remaining within its assessment scope and sphere of influence through sourcing decisions, contractual and other value chain relationships. Software and IT, internet media and telecommunications services depend on datacentres, which in turn rely on semiconductors and other electronic components, each with their own dependencies and impacts on nature.

Dependencies and impacts on nature

The technology and communications sector depends on nature across its value chain, including land, freshwater, the ocean, minerals and metals, and ecosystem services that support a clean, reliable water supply, climate regulation and resilience to natural hazards. These dependencies are particularly relevant for semiconductor fabrication plants (fabs), electronic hardware manufacturing facilities and datacentre operations.

In turn, impacts on nature from the sector arise through land, ocean and freshwater use, minerals and metals extraction, pollution and waste generation associated with semiconductor and other hardware production, datacentre construction and operations. Telecommunications infrastructure can also affect ecosystems through land and ocean use change and possible habitat disturbance when installing cables and towers.

The location, scale and significance of these dependencies and impacts vary across industries, activities and operating contexts. Section 2.2 provides further details of such dependencies and impacts, while all of Section 2 provides guidance on how to apply the LEAP framework to identify and assess dependencies and impacts across the sector.

Nature-related risks and opportunities

These dependencies and impacts on nature can lead to financially material risks for organisations in this sector. For example:

- From May 2024 to March 2025, USD 64 billion of planned US datacentre projects had been blocked or delayed, largely due to local concerns, including concerns about the demand on natural resources and power;[9] and
- The sector's use of water and land is closely linked to energy demand, creating potential financial risks and underscoring the importance of considering climate-nature connections in decision making, investments, strategy and transition planning in this sector.

9 Data Centre Watch (2025) $ 64 billion of data centre projects have been blocked or delayed amid local opposition.

Organisations in the technology and communications sector can also take advantage of nature-related opportunities that benefit both organisations and nature through activities that contribute to positive impacts or mitigate negative impacts on nature. For example:

- Digital technologies, data analytics, sensors, connectivity and software are critical for mapping, monitoring and managing nature-related dependencies and impacts, both within the sector itself and across the wider economy;
- Greater circularity throughout the value chain can decrease costs as well as impacts related to both upstream resource extraction and downstream disposal systems;
- Technology can enable more efficient use of land, water, energy and materials, improve transparency and traceability across value chains, support environmental monitoring and early-warning systems, and facilitate better decision-making by businesses, financial institutions, governments, Indigenous Peoples and Local Communities and other stakeholders;[10] and
- Nature-based solutions such as investments in the conservation and restoration of watersheds, wetlands, floodplains, native forests and coastal ecosystems can increase the resilience of technology and communications infrastructure to increasing risks from drought, flooding and other extreme weather events.

Trade-offs remain as the deployment of emerging technologies to support nature-positive outcomes can also generate negative impacts, including through increased demand for energy, infrastructure and natural resources, as highlighted above.

The significant scale and growth of the technology and communications sector means that companies, insurers, capital providers and other stakeholders are increasingly exposed to these nature-related risks, but also have the potential to capitalise on significant opportunities.[11]

The role of this guidance

This guidance is designed to help organisations in the technology and communications sector to identify and assess the nature-related dependencies, impacts, risks and opportunities across their value chain(s) to inform management and disclosure.[12]

10 Audu, A.J. et al. (2024) The role of digital tools in enhancing environmental monitoring and business efficiency.

11 Kilpalnäinen, K. (2025) Unlocking economic growth through digital infrastructure.

12 See WEF (2025) Nature Positive: Role of the Technology Sector, for a complementary general overview of nature-related risks and opportunities in the technology and communications sector.

2. Sector-specific LEAP assessment guidance

This guidance is a supplement to the TNFD's Guidance on the identification and assessment of nature-related issues: The LEAP approach and should be read in conjunction with that guidance.

Organisations should undertake a robust assessment of nature-related issues to inform their disclosures. The TNFD recommends applying the LEAP approach for this purpose. Section 2 provides additional sector-specific guidance on applying LEAP for technology and communications organisations seeking support in structuring or strengthening their assessment.

Organisations with established processes and strong experience in assessing nature-related issues (including those that want a fast-forward to the metrics) may choose to proceed directly to Section 3 on the sector-specific metrics to disclose.

2.1. Scoping a LEAP assessment

This section provides additional guidance to help technology and communications organisations when scoping a LEAP assessment.

Working hypothesis generation: What are the organisation's activities where there are likely to be material nature-related dependencies, impacts, risks and opportunities?

Value chain considerations when scoping

A technology and communications organisation's dependence and impact on nature may occur in different parts of the value chain (see Figure 3). Material nature-related risks and opportunities may arise from activities under direct operational control, or from activities upstream or downstream.

For example, for an EMS & ODM facility, semiconductors are crucial, even though the production is controlled by their suppliers. Similarly, for software and video streaming companies, datacentres are fundamental to their business model even when they lease them or do not directly control their operations.

When defining the initial assessment boundary, organisations should consider:

- Which activities, services, platforms, products and value chain elements are relevant to their business model;
- Where nature-related dependencies, impacts, risks and opportunities are likely to arise; and

- The operational control, influence, data availability and contractual relationships that can inform the assessment approach, including the level of granularity, data-gathering needs, stakeholder engagement and sequencing of further analysis.

Some large technology and communications organisations may already be required by regulators or investors to assess and disclose material information across global operations and value chains. All organisations may choose to start with a narrower scope to create a manageable starting point for assessment. This could be a small number of high-priority geographic sites or parts of the value chain (product line, platform, service) where material nature-related dependencies, impacts, risks and opportunities are most likely to arise.

The LEAP approach is designed as an iterative process, enabling organisations to begin with the most decision-useful areas. Organisations should progressively expand both the breadth (value chain coverage) and depth (granularity of analysis) of their assessment over time, to include additional value chain activities that are shown to be material to the business and where nature-related dependencies, impacts, risks and opportunities are relevant, in line with increasing data availability, capability and organisational maturity.

Further guidance on assessing upstream and downstream value chains is available in the TNFD guidance on value chains.

Goals and resource alignment:

Given the current level of capacity, skills and data within the organisation and given organisational goals, what are the resource (financial, human and data) considerations and time allocations required and agreed for undertaking an assessment?

Organisations should consider what existing nature, water, biodiversity or climate strategies they have in place, and what data are already collected as part of policies and compliance. This may include data collected from suppliers, regulatory processes and other sustainability and supply chain engagements that may provide information relevant to an assessment using the LEAP approach.

For technology and communications organisations, this may include:

- Asset and location data for semiconductor fabs, electronics and hardware manufacturing sites, datacentres (with some having on-site energy generation facilities) and telecommunication infrastructure such as towers and cables;
- Energy, water, cooling and wastewater data;
- Reuse, recycling and other solid waste management and circularity data;
- Supplier, procurement and due diligence information;
- Data from regulatory processes such as permits and Environmental Impact Assessments; and
- Information from other sustainability, procurement and supply chain engagements.

Key internal stakeholders will also need to be engaged, including but not limited to sustainability, risk, operations, design and engineering, procurement, supply chain and due diligence teams.



2.2. Locate the organisation's interface with nature

This section provides additional guidance to help technology and communications organisations with the Locate phase of the LEAP approach.

L1

L1: Span of the business model and value chain

Guiding questions:

What are our organisation's activities by sector, value chain and geography?
Where are our direct operations?

An organisation in the technology and communications sector should map out its full value chain (see Figure 3).

For some software, IT, internet media, telecommunications or AI-enabled businesses, direct operations may be largely office based, while relevant interfaces with nature may sit elsewhere in the value chain. These may include mineral and metal extraction, semiconductor fabrication, electronics and hardware manufacturing, datacentre construction and operation, telecom towers and cables, as well as downstream reuse, recycling and e-waste management. Organisations should include relevant assets and activities even where they have limited operational control, for example, where they rely on contracted datacentres, shared telecommunications infrastructure or upstream manufacturing capacity.

While datacentres are predominantly located on land, growing demand is driving interest in emerging siting models for datacentres and related infrastructure, including subsea, in polar regions and in space. Where relevant, organisations should consider these siting models and the associated infrastructure when locating activities and identifying interfaces with nature.[13]

The value chain may span multiple countries, biomes and governance contexts, and is often spatially disconnected from an individual organisation's headquarters or core business functions. Activities with potentially moderate or high nature-related dependencies and impacts may therefore sit upstream or downstream from an organisation's direct operations.

When geographically locating assets, organisations should consider GPS coordinates or polygon data within the context of the wider landscape and the organisation's area of influence.

Which activities fall within direct operations (activities and sites over which an organisation has operational or financial control), and which sit upstream or downstream, will vary depending on the organisation's position in the value chain. Table 2 helps organisations

13 Federal Communications Commission (2026) Public Notice – Space Bureau Accepts for Filing Spacex's Application for Orbital Data Centers.

with this value chain mapping in L1 by indicating the types of assets and activities that organisations should seek to locate across their direct operations, as well as upstream and downstream (see Figure 3).

Relevant assets and activities will vary depending on an organisation's business model and position within the value chain.

Table 2: Illustrative assets and activities related to nature across the technology and communications value chain

Industry following value chain	Upstream assets and activities	Assets and activities under direct operational control	Downstream assets and activities
Semiconductors	Assets: • Mining sites and supporting infrastructure Activities: • Raw materials extraction, refining, conversion and processing • Water production, treatment and supply • Energy production and supply	Assets: • Semiconductor fabrication plants (fabs) and supporting infrastructure Activities: • Semiconductor wafer fabrication, assembly, packaging and testing • Water sourcing, ultrapure water (UPW) production, recycling and use • Energy sourcing and use • Chemicals and waste management • Wastewater management, recycling and discharge	Activities: • Semiconductor product use and operation • E-waste collection, treatment, recycling and disposal • Product circularity (e.g. repair, reuse, refurbishment and remanufacturing)
EMS & ODM and hardware	Assets: • Fabrication plants and supporting infrastructure Activities: • Semiconductor wafer fabrication, assembly, packaging and testing • Battery and backup power system manufacturing • Water production, treatment and supply • Energy production and supply	Assets: • Manufacturing facilities and supporting infrastructure Activities: • Semiconductor, chips and electronic component sourcing • EMS & ODM assembly, testing and packaging • Hardware product design and development • Water and energy sourcing and use • Chemicals and waste management • Wastewater management and treatment	Activities: • Hardware product use and operation • E-waste collection, treatment, recycling and disposal • Product circularity (e.g. repair, reuse, refurbishment and remanufacturing)

Industry following value chain	Upstream assets and activities	Assets and activities under direct operational control	Downstream assets and activities
Datacentres (infrastructure)	Assets: • Semiconductor fabs and supporting infrastructure • Manufacturing facilities and supporting infrastructure Activities: • Semiconductor wafer fabrication, assembly, packaging and testing • EMS & ODM assembly, testing and packaging • Hardware product design and development • Battery and backup power system manufacturing • Water production, treatment and supply • Energy production and supply	Assets: • Datacentre facilities and supporting infrastructure Activities: • Hardware procurement, deployment and lifecycle management • Water sourcing and use for cooling systems • Energy sourcing and use • On-site electricity generation and storage (where applicable)	Activities: • Digital, software, IT, internet and media services enabled by datacentre infrastructure • IT equipment lifecycle management (reuse, refurbishment, repair, deployment and recycling) • E-waste collection, treatment, recycling and disposal • Product circularity (e.g. repair, reuse, refurbishment and remanufacturing) • Decommissioning of datacentre infrastructure

Industry following value chain	Upstream assets and activities	Assets and activities under direct operational control	Downstream assets and activities
Telecommunications (infrastructure)	Assets: • Manufacturing facilities and supporting infrastructure Activities: • EMS & ODM assembly, testing and packaging • Fiber optic cable manufacturing and materials processing • Energy production and supply	Assets: • Telecommunications infrastructure (towers, cables, switches, antennas, network equipment) Activities: • Operation and maintenance of network infrastructure • Energy sourcing and use • On-site energy generation and backup power systems (where applicable)	Activities: • Telecommunications services provision and operation (cable, wireless, satellite, broadband and other services) • Customer use of telecommunications services • E-waste collection, treatment, recycling and disposal • Product circularity (e.g. repair, reuse, refurbishment and remanufacturing) • Decommissioning of telecom infrastructure
Software and IT services (for datacentre infrastructure, formally part of Software and IT SICS® category, see above)	Assets: • Datacentre facilities and supporting infrastructure Activities: • Datacentre operations including water and energy sourcing and use for cooling systems	Assets: • Software and IT operational environments (e.g. cloud regions, on-premises environments) • Digital platforms, applications and services Activities: • Software development, deployment and operation • Data processing, hosting and managed IT services • Application performance and resource optimisation (e.g. software efficiency, architecture, scaling behaviour) • AI model training (where applicable)	Activities: • Software and IT services deployment and use • Customer use of digital products and platforms • IT equipment lifecycle management (reuse, refurbishment, repair, deployment and recycling) • E-waste collection, treatment, recycling and disposal • Product circularity (e.g. repair, reuse, refurbishment and remanufacturing)

Industry following value chain	Upstream assets and activities	Assets and activities under direct operational control	Downstream assets and activities
Internet Media and Services (TC-IM) (for datacentre and telecommunications infrastructure, see above)	Assets: • Datacentre facilities and supporting infrastructure • Telecommunications infrastructure Activities: • Datacentre operations including water and energy sourcing and use for cooling systems • Telecommunications network operations including energy sourcing and use	Assets: • Software and IT operational environments • Digital platforms, applications and services Activities: • Software development, deployment and operation • Digital content and service delivery • Data processing, hosting and managed IT services • Application performance and resource optimisation • AI model training (where applicable)	Activities: • Internet and media services deployment and use • Customer use of digital platforms and applications
Telecommunications services	Assets: • Telecommunications infrastructure • Datacentre facilities supporting telecom services (where applicable) Activities: • Telecommunications network operations including energy sourcing and use • Datacentre operations including water and energy sourcing and use for cooling systems (where applicable)	Assets: • Operation and maintenance of network infrastructure Activities: • Telecommunications services provision and operation • Datacentre operations' supporting network services (where applicable)	Activities: • Telecommunications services (cable, wireless, satellite and other services) deployment and use • Customer use of telecommunications services • E-waste collection, treatment, recycling and disposal • Product circularity (e.g. repair, reuse, refurbishment and remanufacturing)

L2

L2: Dependency and impact screening

Guiding question:

Which of these sectors, value chains and direct operations are associated with potentially moderate and high dependencies and impacts on nature?

In L2, organisations consider which of the activities and assets identified in L1 (after analysing the value chain) are associated with potentially moderate and high dependencies and impacts on nature. This is a first phase of screening, with more detailed measurement in the Evaluate phase.

Dependencies on nature

The resilience of the sector depends on nature. Land is needed for resource extraction, semiconductor fabs, manufacturing facilities, datacentre campuses and, increasingly, the energy infrastructure for energy-intensive operations such as datacentres. Healthy watersheds and groundwater sources are crucial for a consistent and clean water supply, particularly during droughts. Where operations are located in water-scarce regions, this dependency creates exposure to physical risks such as water scarcity, declining water quality and operational disruption.[14]

Healthy ecosystems can also reduce hazard exposure and support operational resilience. Intact watersheds, wetlands, floodplains and coastal ecosystems can help buffer against droughts and floods from extreme rainfall events[15] that can disrupt fab operations, datacentre cooling, datacentre and telecommunications infrastructure and wider supply chains.

Impacts on nature

Impacts on nature from economic activities in the sector are driven by land use change, water use, minerals and metals extraction, and pollution, including air, soil and water pollution across the value chain and significant waste generation. The impacts mentioned below appear primarily in field-based direct operations (semiconductor fabs, manufacturing facilities, and datacentre and telecommunications infrastructure). Sub-sectors upstream (and their customers), such as software and IT companies and those providing internet and media and telecommunication services, should consider such impacts as part of their upstream value chain.

- **Land, freshwater and ocean use change:** An estimated 160 square kilometres of ‘powered land’ will be needed to support projected datacentre growth over the next five years.[16] This represents only part of the wider land footprint: the associated land footprint of electricity infrastructure is projected to exceed 14,500 square kilometers

14 AXA (2023) Why Should Water Stewardship Go Beyond ESG Strategy?

15 UNDP (2025) Nature for Disaster Risk Register.

16 Hines Research (2025) Power Play: The Emerging Powered Land Opportunity.

by 2030.[17] An individual datacentre may require two to three square kilometres of land for about one gigawatt of power, with the energy use also requiring significant water resources.[18] Meanwhile, telecommunications towers and subsea cables require space across land and seascapes and have the potential to disturb and fragment habitats during construction.

- **Water use:** Semiconductor manufacturing is highly water intensive and an average single chip fabrication plant can use around 38 million litres of ultrapure water (UPW) per day. Though water use can vary significantly, a typical datacentre that relies on water for cooling and energy production can use 25 million to 770 million litres of water per year (depending on size),[19] while hyperscale facilities may exceed 2 billion litres annually.[20] Datacentre water consumption also includes indirect consumption through the electricity generated and the water and wastewater services used by the facility. This can represent up to 75% of total water consumption, depending on the electricity generation mix.[21] The rapid growth of AI is expected to result in an estimated 731 to 1,125 million cubic metres of water consumption annually by 2030 – equivalent to the yearly household water use of 6 to 10 million Americans.[22]

  Overexploitation of water basins can affect freshwater ecosystem health. This in turn can affect Indigenous Peoples and local communities, farmers and other stakeholders that rely on healthy water basins for drinking water, irrigation of crops and other uses. This can also affect the technology and communications sector itself, given its own dependency on water, including UPW for semiconductor manufacturing and cooling for datacentres.

- **Metals and critical minerals use:** Semiconductor and electronics manufacturing uses a wide range of metals and critical minerals as inputs. This links downstream industries to upstream mining-related land use, pollution and biodiversity loss, and the dependencies on nature of the metals and mining sector, such as water supply and natural hazard regulation.[23]
- **Other materials use:** Technology and communications infrastructure also relies on non-metallic materials, including polymers derived from crude oil and natural gas for cable coatings and electronic components, and sand and other construction materials. Their

17 Aczel, M. et al. (2026) Environmental Cost of AI's Energy Use: Carbon, Water and Land Footprints. United Nations University Institute for Water, Environment and Health.

18 WEF (2025) Nature Positive: Role of the Technology Sector.

19 Ceres (2025) Drained by Data: The Cumulative Impact of Data Centers on Regional Water Stress; Ceres (2026) Water Behind the Watts: The Hidden Risk of Powering Data Centers.

20 WEF (2025) Nature Positive: Role of the Technology Sector.

21 NatureFinance (2025) Navigating AI's Thirst in a Water-Scarce World: A Governance Agenda for AI and the Environment.

22 Nutt, D. (2025) 'Roadmap' shows the environmental impact of AI data center boom.

23 TNFD (2024) Additional sector guidance – Metals and mining.

extraction and production can contribute to land use change, water use, pollution and other impacts on nature.

- **Waste:** Hardware and electronics manufacturing generates substantial waste streams, including hazardous waste. By 2030, global e-waste is expected to reach 82 billion kilogrammes, an estimated increase of 30% compared with 2024.[24] Limited recycling rates and waste management practices can result in pollution of air, soils and water. Only 20% to 22% of e-waste is documented as properly collected and recycled.[24]

Materiality tables

Table 3 lists ecosystem services that different sub-sectors within technology and communications typically depend on. Table 4 lists the typical impact drivers. An organisation can use these tables as a first filter to screen sectors and activities across direct operations and value chains for potentially moderate and high dependencies and impacts on nature.

The materiality ratings in Tables 3 and 4 use ENCORE as a foundational framework. In ENCORE, activities are mapped to the International Standard Industrial Classification (ISIC). Information on which ISIC groups were included under each SASB category is provided in Annex 2. Where activities within the scope of this guidance did not directly map back to any ISIC group, adjustments were made to better reflect the nature-related dependencies and impacts associated with activities in the sectors and industries within scope. These adjustments were informed by academic literature and relevant technical reports.

Each column has industry-average ratings for the direct operations of the industry at the top of that column. Following the steps in L1 and informing later stages of LEAP, every organisation should consider ratings in the column for its own industry and relevant upstream and downstream activities. For instance, an organisation that provides video streaming services (part of Internet & Media) would look upstream in its value chain, and consider the dependencies and impacts of datacentre operations, the dependencies and impacts of the hardware in the server racks, and the semiconductors within them.

Technology and communications organisations should refer to other TNFD sector guidance documents and the ENCORE materiality tables included when screening activities in related value chain sectors (e.g. metals and mining).

While the materiality ratings set out in Tables 3 and 4 can support an initial assessment, they are based on industry averages, are non-exhaustive and may not fully capture all industry- and context-specific considerations. Organisations are encouraged to use the tables as a starting point when screening for material dependencies and impacts, while adapting and refining their assessment in subsequent components of the LEAP approach. Organisations can also consider the WWF Risk Filter and the SBTN High Impact Commodities List and Materiality Screening Tool.

24 Geneva Environment Network (2025) The Growing Environmental Risks of E-Waste – Geneva Environment Network; Cordelis, P. et al. (2024) The Global E-waste Monitor 2024.

Table 3: Materiality ratings of ecosystem services the technology and communications sector typically depends on

Ecosystem services		Manufacturing, design and sales of electronic equipment (see other columns for upstream/downstream)			Physical infrastructure (see other columns for upstream/ downstream)		Digital products & services (see other columns for upstream/downstream)		
		Semi-conductors	EMS & ODM	Hardware (including consumer electronics)	Datacentre infrastructure	Telecom infrastructure	Telecom Services (for infrastructure see other columns)	Software & IT Services	Internet Media & Services
Provisioning services	Water supply	VH	M	L	VH/H*	VL	VL	VL/L	VL/L
	Genetic material services	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Biomass provisioning services	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Other provisioning services – Animal-based energy	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regulating and maintenance services	Solid waste remediation	L	M	M	M	N/A	N/A	L	L
	Soil and sediment retention services	L	L	L	L	M	L	VL	VL
	Water purification services	VH	M	M	VH/H	N/A	N/A	N/A	N/A

Ecosystem services		Manufacturing, design and sales of electronic equipment (see other columns for upstream/downstream)			Physical infrastructure (see other columns for upstream/ downstream)		Digital products & services (see other columns for upstream/downstream)		
		Semi-conductors	EMS & ODM	Hardware (including consumer electronics)	Datacentre infrastructure	Telecom infrastructure	Telecom Services	Software & IT Services	Internet Media & Services
							(for infrastructure see other columns)		
Regulating and maintenance services	Soil quality regulation services	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Other regulating and maintenance service – Dilution by atmosphere and ecosystems	L	L	L	H	N/A	N/A	N/A	N/A
	Biological control services	N/A	N/A	VL	N/A	N/A	N/A	N/A	N/A
	Air filtration services	VL	VL	VL	M	N/A	N/A	L	L
	Flood mitigation services	M	M	M	H	H	L	VL/L	VL
	Global climate regulation services	VL/L	VL	VL	VL	M	VL	VL	VL
	Nursery population and habitat maintenance services	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Ecosystem services		Manufacturing, design and sales of electronic equipment (see other columns for upstream/downstream)			Physical infrastructure (see other columns for upstream/ downstream)		Digital products & services (see other columns for upstream/downstream)		
		Semi-conductors	EMS & ODM	Hardware (including consumer electronics)	Datacentre infrastructure	Telecom infrastructure	Telecom Services (for infrastructure see other columns)	Software & IT Services	Internet Media & Services
Regulating and maintenance services	Noise attenuation services	VL	VL	N/A	N/A	N/A	N/A	N/A	N/A
	Other regulating and maintenance service – Mediation of sensory impacts (other than noise)	VL	VL	N/A	N/A	N/A	N/A	N/A	N/A
	Local (micro and meso) climate regulation services	L	L	L	H	L	L	L	L
	Pollination services	N/A	N/A	NA	N/A	N/A	N/A	N/A	N/A
	Storm mitigation services	M	M	M	VL	H	L	VL/L	VL
	Water flow regulation services	M/H	M	M	M	M	L	VL/L	VL/L
	Rainfall pattern regulation services (at sub-continental scale)	VL/L	M	VL	VL	VL	VL	VL	VL

Ecosystem services		Manufacturing, design and sales of electronic equipment (see other columns for upstream/downstream)			Physical infrastructure (see other columns for upstream/downstream)		Digital products & services (see other columns for upstream/downstream)		
		Semi-conductors	EMS & ODM	Hardware (including consumer electronics)	Datacentre infrastructure	Telecom infrastructure	Telecom Services	Software & IT Services	Internet Media & Services
							(for infrastructure see other columns)		
Cultural services	Recreation-related services	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Visual amenity service	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Education, scientific and research services	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Spiritual, artistic and symbolic services	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

*Dependency on water supply may vary across direct and upstream operations, influenced by cooling technology, facility design, climatic conditions and electricity sources.

Source: ENCORE Partners (Global Canopy, UNEP FI and UNEP-WCMC) (2024). ENCORE: Exploring Natural Capital Opportunities, Risks and Exposure. Cambridge, UK: the ENCORE Partners. DOI: https://doi.org/10.34892/dz3x-y059. The ENCORE data displayed here is made available to readers under a CC BY-SA 4.0 licence (https://creativecommons.org/licenses/by-sa/4.0/deed.en); World Economic Forum (2025) Nature Positive: Role of the Technology Sector and additional research by TNFD.

Table 4: Materiality ratings for impact drivers typically relevant for the technology and communications sector

Drivers of nature change	Impact drivers[25]	Manufacturing, design and sales of electronic equipment (see other columns for upstream/downstream)			Physical infrastructure (see other columns for upstream/ downstream)		Digital products & services (see other columns for upstream/downstream)		
		Semi-conductors	EMS & ODM	Hardware (including consumer electronics)	Datacentre infrastructure	Telecom infrastructure	Telecom Services (for infrastructure see other columns)	Software & IT Services	Internet Media & Services
Land/ Freshwater/ Ocean Use Change	Area of land use	L	L	L	H/VH	M/H	VL	L	L
	Area of freshwater use	N/A	N/A	N/A	N/A	L	L	N/A	N/A
	Area of seabed use	N/A	N/A	N/A	N/A	M	N/A	N/A	N/A
Climate Change	Emission of GHG	H	M/H	M/H	H/VH	H/VH	L	L/M	L
Pollution/ Pollution Removal	Emissions of non-GHG air pollutants	H	L/M	L/M	M/H	VL	VL	VL/L	L
	Emissions of toxic pollutants to water and soil	H	H	H	H	H	L	L	L
	Emissions of nutrient pollutants to water and soil	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

25 Impact drivers are referred to as 'pressures' in ENCORE.

Drivers of nature change	Impact drivers[25]	Manufacturing, design and sales of electronic equipment (see other columns for upstream/downstream)			Physical infrastructure (see other columns for upstream/downstream)		Digital products & services (see other columns for upstream/downstream)		
		Semi-conductors	EMS & ODM	Hardware (including consumer electronics)	Datacentre infrastructure	Telecom infrastructure	Telecom Services	Software & IT Services	Internet Media & Services
							(for infrastructure see other columns)		
Pollution/ Pollution Removal	Generation and release of solid waste	M/H	M/H	H	M/H	M	VL	VL	VL/L
	Disturbances	L/M	M	M	M/H	H		L	VL/L
Resource Use/ Replenishment	Other biotic resource extraction only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Other abiotic resource extraction	H/VH	VH	VH	H/VH	M/H	N/A	N/A	N/A
	Volume of water use	H/VH	H	H	H/VH*	VL	VL	L	L/M
Invasive Alien Species Introduction/ Removal	Introduction of invasive species	N/A	N/A	ND	N/A	N/A	N/A	N/A	N/A

*Volume of water use may vary depending on cooling technology, facility design and climatic conditions. Significant water use impacts may occur upstream, particularly those associated with electricity generation and physical infrastructure.

Source: ENCORE Partners (Global Canopy, UNEP FI and UNEP-WCMC) (2024). ENCORE: Exploring Natural Capital Opportunities, Risks and Exposure. Cambridge, UK: the ENCORE Partners. DOI: https://doi.org/10.34892/dz3x-y059. The ENCORE data displayed here is made available to readers under a CC BY-SA 4.0 licence (https://creativecommons.org/licenses/by-sa/4.0/deed.en); World Economic Forum (2025) Nature Positive: Role of the Technology Sector and additional research by TNFD.

25 Impact drivers are referred to as 'pressures' in ENCORE.

Using data centre archetypes to improve initial screening

Actual dependencies and impacts will vary across organisational and geographical contexts. Organisations can use other more targeted or localised datasets to support screening where available or refer to locally defined or international standards that set out nature-related dependencies and impacts of the technology and communications sector.

As an example, for those that own, operate or depend strongly on datacentres in their business model, the WEF's datacentre archetypes summary (Figure 5) may be useful both in L2 and the Evaluate phase.

Figure 5: A summary of datacentre (DC) archetypes and implications

Archetype	Cooling degree days*	Water stress level	Renewable energy share	Description	Implication
High cooling need, versatile cooling options	High/very high	Low/medium	High/very high	Regions with material cooling needs that have access to renewable power and low water stress	DCs can optimise efficiency through a variety of energy sources and cooling methods
High cooling need, prioritise renewable power	High/very high	High/very high	High/very high	Regions with material cooling needs that have access to renewable power but are water stressed	DCs may look to renewable power to maximise efficiency and reduce water use
High cooling need, incorporate water cooling	High/very high	Low/medium	Low/medium	Regions with material cooling needs that have limited access to renewable power but low water stress	DCs focus on energy efficiency focusing less on water usage in the near term
High cooling need, constrained on cooling options	High/very high	High/very high	Low/medium	Regions with material cooling needs that have limited access to renewable power and high water stress	DCs may require innovative solutions (e.g. on site water recycling, SMRs for power)** to optimise
Low cooling need, optionality in cooling	High/very high	Low/medium	High/very high / Low/medium	Regions with low cooling needs that have access to renewable power or have low water stress or both	DCs optimise operations based on the available resources with limited impact
Low cooling need, constrained on cooling options	High/very high	High/very high	Low/medium	Regions with low cooling needs that have limited access to renewable power and high water stress	DCs may look to low-impact cooling methods (e.g. free air cooling) when possible

Low/medium High/very high

SAMPLE IMPLICATION

Dallas

Dallas has significant cooling needs, but optionality given moderate water stress and high renewable energy share.

SAMPLE IMPLICATION

United Arab Emirates

DCs in the United Arab Emirates may seek innovative solutions (e.g. water recycling) to manage cooling requirements.

- High cooling need, but versatile
- High cooling need, renewable power
- High cooling need, water cooling
- High cooling need, constrained
- Low cooling need, optionality
- Low cooling need, constrained
- Very high/high solar potential
- Very high/high wind potential
- Very high/high geothermal potential

Notes: *Cooling degree days are used to indicate cooling requirements. The metric shows how far above 15.5°C the daily average temperature is over the course of two years. See additional methodology in Appendix B of WEF (2025).
**SMR = small modular reactor.

Source: World Economic Forum (2025) Nature Positive: Role of the Technology Sector

These archetypes can support initial screening by helping organisations identify which dependencies and impacts may be most relevant across different datacentre operating contexts. Organisations can compare characteristics such as cooling demand, water stress and renewable energy share to determine which archetype best reflects a particular site or group of sites and use this to inform where further assessment may be needed.

For example, an organisation that operates datacentres across several regions could use the archetypes to distinguish sites with high cooling needs ánd high water stress from those with lower cooling needs or greater flexibility in cooling options. This can help indicate where water-related dependencies and impacts may require greater attention, while also highlighting trade-offs between water use, cooling technology and energy sources.

The archetypes are intended as an initial screening tool, rather than a substitute for site-level assessment. Subsequent phases of LEAP add more granularity to dependency and impact assessment by considering local conditions, operational practices, and the trade-offs between cooling, water use and energy sources (water-based cooling being only one of several cooling techniques). These factors and their implications should be assessed at the site level.

L3

L3: Interface with nature

Guiding questions:

Where are the sectors, value chains and direct operations with potentially moderate and high dependencies and impacts located? Which biomes and specific ecosystems do our direct operations and moderate and high dependency and impact value chains and sectors interface with?

Based on the value chain assessment in L1 and the materiality screening in L2, an organisation can determine which biomes and ecosystems these assets and activities interface with.

Organisations in the technology and communications sector typically interface with the following biomes:

Land – for any fabs, manufacturing and processing sites, datacentre or telecom infrastructure.

- Temperate boreal forests and woodlands (T2);
- Shrublands and shrubby woodlands (T3);
- Savannas and grasslands (T4); and
- Deserts and semi-deserts (T5).

Freshwater – especially relevant for infrastructure requiring ample access to freshwater.

- Rivers and streams (F1);
- Lakes (F2);
- Artificial wetlands (F3);
- Vegetated wetlands (TF1); and
- Subterranean freshwaters (SF1).

Ocean – especially relevant for the underwater cables part of telecommunications infrastructure.

- Marine shelf (M1);
- Deep sea floors (M3);
- Shoreline systems (MT1);
- Maritime vegetation (MT2);
- Coastal inlets and lagoons (FM1); and
- Brackish tidal systems (MFT1).

This list can be considered as a reference. However, organisations should review all applicable biomes connected to their specific interfaces across the value chains and associated activities where significant dependencies and impacts on those biomes exist.

The following are illustrative examples of technology and communications sector organisations' interfaces with biomes and ecosystems:

- Semiconductor fabs may be located in or near water basins, in biomes such as rivers and streams (F1), lakes (F2) and subterranean freshwaters (SF1);
- Datacentres that depend on water for energy production and/or cooling may be located in land-based biomes but can also be in or near freshwater biomes such as rivers and streams (F1), lakes (F2) and vegetated wetlands (TF1);
- Telecommunication towers may be located in a wide range of land-based biomes, such as temperate boreal forests and woodlands (T2), shrublands and shrubby woodlands (T3), savannas and grasslands (T4), and deserts and semi-deserts (T5); and
- For telecommunications organisations, cables under seabeds may be located in ocean biomes, such as the marine shelf (M1) and deep-sea floors (M3).

Organisations should also consider how the type and spatial pattern of assets may affect the interface with ecosystems. Concentrated assets, such as datacentres, manufacturing facilities and telecommunications network hubs may have localised but potentially intensive dependencies and impacts linked to land use, water, energy and waste. Distributed assets, such as towers, cables and other network infrastructure, may have smaller individual footprints but can create cumulative impacts across landscapes, basins or seascapes,

including through construction, maintenance, habitat disturbance and the potential spread of invasive alien species.

An organisation can identify the ecosystems in which raw materials are mined by referring to datasets such as Tang et al.[26] to better understand the origin of metals and minerals procured and identify the biomes and ecosystems they are located in. Organisations may refer to L3 in the LEAP guidance and to the TNFD biome guidance for more definitions and further guidance when analysing their interfaces with biomes and ecosystems.

L4

L4: Interface with sensitive locations

Guiding questions:

For our organisation's activities in moderate and high dependency and impact value chains and sectors, which of these are in ecologically sensitive locations? And which of our direct operations are in sensitive locations?

Building on the assessment in L1 to L3, organisations assess whether identified activities and assets (direct operations and in value chain) are geographically located in ecologically sensitive locations. Sensitive locations are defined by TNFD as:

- Areas important for biodiversity, including species; and/or
- Areas of high ecosystem integrity; and/or
- Areas of rapid decline in ecosystem integrity; and/or
- Areas of high physical water risks; and/or
- Areas of importance for ecosystem service provision, including benefits to Indigenous Peoples, Local Communities and stakeholders.

Organisations can refer to the criteria for sensitive locations in Table 4 of the TNFD LEAP approach. Sensitive locations may extend well beyond the footprint of an individual asset. Semiconductor fabs and some EMS & ODM facilities can depend heavily on local water resources and may affect the very water basins they depend on through abstraction, wastewater discharge and chemical use. Datacentres may create similar location-specific pressures where water is used for cooling or where electricity supply has a water footprint. Organisations should therefore consider not only the sensitivity of the site itself, but also the wider basin and the needs of other users (including Indigenous Peoples, local communities, farmers and other stakeholders) that depend on the same ecosystem services.

Telecommunications infrastructure has a different spatial pattern of interaction with sensitive locations. Towers, terrestrial cable routes and subsea cables can cross or connect multiple ecosystems, including wildlife corridors and other areas important for habitat connectivity. Relevant interfaces may occur along cable routes and at landing or connection points and not just at individual sites. The area of assessment should therefore reflect how each

26 Tang, L. and Werner, T.T. (2023) Global mining footprint mapped from high-resolution satellite imagery.

activity interacts with the surrounding landscape, basin or seascape, including ecological connections and pathways through which impacts may occur.

Organisations can use asset-level spatial data, such as site polygons, cable routes, tower locations, water abstraction points and discharge points, and overlay these with spatial data on sensitive locations. For example, a fab or datacentre can be assessed in relation to basin-level water stress and related freshwater ecosystems, while a subsea cable route can be checked against marine habitats along the route and at landing points. This can help identify where activities or assets interface with sensitive locations and where more detailed assessment is needed.

Sources of further information, datasets and tools include:

- WWF Risk Filter (containing both the Biodiversity Risk Filter and the Water Risk Filter);
- WRI Aqueduct;
- EMODNet (dataset on subsea telecommunication cables in European waters);
- Global Lakes and Wetlands Database;
- Global Nature Watch (formerly the Global Forest Watch, since widened in scope);
- Integrated Biodiversity Assessment Tool (IBAT); and
- UN Biodiversity Lab.

Further tools and guidance are provided in the TNFD guidance on the LEAP approach, the TNFD Tools Catalogue, the UNEP-WCMC Nature Tools Compass and SBTN's Step 1 Toolbox.



2.3. Evaluate dependencies and impacts on nature

This section provides additional guidance to help technology and communications organisations with the Evaluate phase of the LEAP approach.

E1

E1: Identification of environmental assets, ecosystem services and impact drivers

Guiding questions:

What are the sub-sectors, business processes or activities to be analysed? What environmental assets, ecosystem services and impact drivers are associated with these sub-sectors, business processes, activities and assessment locations?

Guidance for components E1 and E2 is provided together under E2.

E2

E2: Identification of dependencies and impacts

Guiding question:

What are our dependencies and impacts on nature?

In the Evaluate phase, organisations move beyond the global averages used for screening in L2 to consider dependencies and impacts in the context of the organisation's actual operations and specific locations. Dependencies and impacts on nature arise across technology and communications organisations' direct operations and upstream and downstream value chains.

Impacts may extend beyond the site, such as pollutants carried downstream through river systems to the ocean. This may include water withdrawal or discharge points (where they link to wider water basins), site-specific energy generation or backup power infrastructure, cooling and wastewater systems, access routes and cable corridors.

Tables 5 and 6 provide illustrative dependency and impact pathways for the technology and communications sector. Organisations should refer to TNFD additional sector guidance for dependencies and impacts upstream and downstream where relevant (e.g. metals and mining, engineering, construction and real estate, construction materials, and electric utilities and power generators).

Dependencies on nature

Table 5 provides examples of dependency pathways relevant to technology and communications. It is not intended to be comprehensive and an organisation will need to consider if these or other examples apply to its particular context.

Table 5: Examples of dependency pathways in the technology and communications sector, by business activity

Business activity	Environmental assets and ecosystem services depended on	Description of dependencies
Semiconductors		
Raw materials procurement	**Environmental assets** • Underwater mineral and energy resources • Marine ecosystems • Land • Terrestrial and subterranean terrestrial ecosystems • Mineral and energy resources • Water resources • Freshwater ecosystems **Ecosystem services** • Soil and sediment retention • Flood mitigation • Water flow regulation • Rainfall pattern regulation • Local and global climate regulation • Storm mitigation	The semiconductors industry depends on a wide range of components: minerals (including but not limited to silica sand, silicon and indium),[27] rare earth elements (REEs) and critical and precious metals (gold, silver, copper, aluminium, platinum, palladium). As such, they depend on natural processes that create these resources. The semiconductors industry also relies on regulating and maintenance ecosystem services for the operation of mining activities to supply the sector's components (e.g. chips). Ecosystem services that support mining infrastructure and its operations include effective flood mitigation, rainfall pattern regulation, storm mitigation to reduce risks such as excessive rainfall and flooding and local climate regulation. More information on dependencies is provided in TNFD's Metals and Mining sector guidance.

27 Goswami, O. (2023) Chipping in: Critical minerals for semiconductor manufacturing in the US.

Business activity	Environmental assets and ecosystem services depended on	Description of dependencies
Semiconductor wafer fabrication, assembly, packaging and testing	**Environmental assets** • Water resources • Freshwater ecosystems **Ecosystem services** • Water supply • Water flow regulation • Water purification • Rainfall pattern regulation	Water is critical for semiconductor manufacturing and for operating facilities. It is notably used for air purification, temperature control, water purification and cooling systems in very high volumes. As a direct physical input, organisations are dependent on sufficient freshwater from both surface water and groundwater for their manufacturing processes. The availability of water, in turn, depends on reliable rainfall and the wider hydrological cycle to recharge water bodies. For example, in East Asia, the absence of rainfall has reduced capacity to capture and store water, contributing to drought conditions that have forced manufacturers to rely on hundreds of water trucks to maintain their operations.[28]
EMS & ODM and Hardware		
Raw materials and semiconductors procurement	Refer to the semiconductors section for environmental assets and ecosystem services.	In addition to raw materials for semiconductor production, EMS & ODM and hardware depend on battery materials, including lithium, cobalt and graphite.

28 Roussilhe, G. et al. (2024) From silicon shield to carbon lock- in? The environmental footprint of electronic components manufacturing in Taiwan (2015–2020).

Business activity	Environmental assets and ecosystem services depended on	Description of dependencies
EMS & ODM assembly, testing, packaging Hardware product design and development	**Environmental assets** • Water resources • Freshwater ecosystems **Ecosystem services** • Water supply • Water purification • Water flow regulation	EMS & ODM and hardware manufacturing, assembly and testing depend on reliable supplies of water of suitable quality for process cleaning and rinsing, cooling, humidity control and facility operations. This, in turn, depends on freshwater resources and ecosystem services that support water supply, natural purification and flow regulation. The level of dependence varies by product and production process, with component manufacturing generally more water intensive than final assembly and testing.
	Environmental assets • Land • Terrestrial ecosystems • Freshwater ecosystems • Marine and coastal ecosystems (e.g. mangroves protecting coastline) **Ecosystem services** • Flood mitigation • Rainfall pattern regulation • Storm mitigation	An example that illustrates the industry's dependence on services like flood mitigation, rainfall pattern regulation and storm mitigation is the flooding in Thailand that severely disrupted electronics manufacturing and caused global shortages, leading to significant price spikes for electronic components.[29]

29 Ploy Ten, K. and Kim, C.R. (2011) Thai floods batter global electronics, auto supply chains.

Business activity	Environmental assets and ecosystem services depended on	Description of dependencies
Datacentres (infrastructure)		
Construction of datacentre facilities and supporting infrastructure	**Environmental assets** • Land • Terrestrial (land-based) ecosystems • Subterranean terrestrial ecosystems **Ecosystem services** • Rainfall pattern regulation • Flood mitigation • Soil and sediment retention • Storm mitigation	The construction of datacentres requires land and various ecosystem services to operate. Ecosystems surrounding datacentres provide regulating and maintenance services that protect operations from natural hazards such as floods and storms and help to stabilise the surrounding soil, supporting operations and reducing the risk of landslides.
Hardware procurement	Refer to semiconductors, EMS & ODM and hardware sections for more information.	

Business activity	Environmental assets and ecosystem services depended on	Description of dependencies
Water sourcing and use for electricity generation	**Environmental assets** • Water resources • Freshwater ecosystems **Ecosystem services** • Water supply • Water flow regulation • Water purification	Datacentres may depend indirectly on water resources to produce electricity that allows them to run continuously. Reliance on water is generally when the source of electricity is fossil fuels instead of renewable sources like solar or wind. Datacentres rely on the hydrological cycle to recharge groundwater sources, regulate river flows and maintain surface water flows, notably groundwater and lake water tables.[30]
Water sourcing and use for cooling systems	**Environmental assets** • Water resources • Freshwater ecosystems **Ecosystem services** • Water supply • Water flow regulation • Water purification	Datacentres using water-based cooling systems require water for cooling and/or humidification. They depend on water supply and healthy water basins, which support groundwater recharge and help regulate river and lake levels and flows.

30 Ceres (2025) Drained by Data: The Cumulative Impact of Data Centers on Regional Water Stress; Ceres (2026) Water Behind the Watts: The Hidden Risk of Powering Data Centers.

Business activity	Environmental assets and ecosystem services depended on	Description of dependencies
Telecommunications (infrastructure)		
Construction of telecommunications infrastructure (towers, cables, switches, antennas, network equipment)	**Environmental assets** • Land • Terrestrial (land-based) ecosystems • Subterranean terrestrial ecosystems • Marine (ocean) ecosystems • Subterranean marine and coastal ecosystems **Ecosystem services** • Rain pattern regulation • Flood mitigation • Soil and sediment retention • Storm mitigation	Building telecommunication infrastructure requires land for facilities, including the installation of towers, cables for transmission lines and ground stations associated with broadcasting satellites. Ecosystems surrounding telecommunications infrastructure provide regulating and maintenance services that protect infrastructure from natural hazards such as floods and storms and help to stabilise the surrounding soil, reducing the risk of landslides. Submarine cable systems depend on the stability of the seafloor along their entire route, where sediment regimes, seabed stability, ocean conditions and climatic patterns are all contributing factors.[31] To ensure cable protection and long-term stability, their placement must intentionally avoid sensitive habitats based on accurate environmental information, including the distribution of benthic habitats, the nature of seafloor substrates and sediment dynamics.[32]

31 Bricheno, L. et al. (2024) The diversity, frequency and severity of natural hazard impacts on subsea telecommunications networks.

32 UNEP-WCMC (2025) Submarine cables and marine biodiversity.

Business activity	Environmental assets and ecosystem services depended on	Description of dependencies
Software IT services, Internet & Media services and Telecommunications services		
Use of datacentre facilities and supporting infrastructure	Refer to datacentres' infrastructure section for more information.	
Software development, deployment and operation	**Environmental assets** • Land • Water resources • Freshwater ecosystems **Ecosystem services** • Water supply • Water flow regulation • Water purification	Software efficiency, architecture and scaling behaviour depend on energy and water based on the cooling demand of datacentres, which in turn is influenced by computational intensity and regional deployment choices. For more information on associated dependencies, refer to the datacentres section.

Table 6 provides examples of impact pathways that are relevant to technology and communications organisations. It is not intended to be comprehensive and an organisation will need to consider if and how the examples apply to its particular context. Impacts can also be positive if a technology and communications organisation improves the state of nature in its direct operations or across the value chain.

Table 6: Examples of impact pathways for the technology and communications sector, by activity

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
Semiconductors			
Raw materials sourcing	Resource use/ replenishment and others	**Environmental assets** • Mineral and energy resource • Underwater mineral and energy resource **Ecosystem services** • Potentially all	The growth in this industry has driven demand for minerals and metals, including those considered critical due to supply risks. Global recycling rates for many specialty elements from e-waste remain very low, often below 1%,[33] which means primary extraction continues to be the main source. Where relevant, organisations should also consider indirect water use associated with mining. Refer to the TNFD Metals and Mining sector guidance for more information.
Semiconductor wafer fabrication, assembly, packaging and testing	Resource use/ replenishment	**Environmental assets** • Water resources • Freshwater ecosystems **Ecosystem services** • Water supply • Water flow regulation	A large semiconductor fabrication plant (fab) uses up to 38 million litres of water per day, equivalent to the daily water consumption of 300,000 people in Germany. Water is primarily used for ultrapure water (UPW) production, a complex process that involves multi-stage treatments. The environmental impacts of this water consumption can be severe. Water reuse and recycling are common on-site practices in semiconductor fabs in certain regions of the world.[34]

33 Buechler, D. T. et al. (2020) Comprehensive elemental analysis of consumer electronic devices: rare earth, precious, and critical elements.

34 Hess, J.C. (2024) Chip Production's Ecological Production: Mapping Climate and Environmental Impact.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
Semiconductor wafer fabrication, assembly, packaging and testing	Pollution/pollution removal	**Environmental assets** • Atmospheric systems **Ecosystem services** • Local (micro and meso) climate regulation • Global climate regulation • Air filtration	Semiconductor manufacturing generates a variety of air pollutants beyond greenhouse gases, notably Volatile Organic Compounds (VOCs), acidic gases and particulate matter (primarily PM10 and smaller). Due to the diverse production technologies used in the semiconductor industry, the source profiles of different pollutants differ significantly.[35]
Semiconductor wafer fabrication, assembly, packaging and testing	Pollution/pollution removal	**Environmental assets** • Land • Terrestrial (land based) ecosystems • Freshwater ecosystems • Marine (ocean) ecosystems **Ecosystem services** • Water supply • Water purification • Soil quality regulation • Solid waste remediation	Semiconductor production leads to a significant amount of hazardous and non-hazardous waste. Waste typically consists of heavy metals, solvents and corrosive compounds in both solid and liquid forms. Solid waste is often processed at designated hazardous waste disposal facilities. In some cases, the heavy metals can be recycled. Release of waste without proper treatment can cause severe environmental impacts.[36]

35 Yin, Y., and Yang, Y. (2025) Sustainable Transition of the Global Semiconductor Industry: Challenges, Strategies, and Future Directions.

36 Veolia (2022) The CHIPS Act: Balancing Manufacturing Capacity and Waste Generation.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
Semiconductor wafer fabrication, assembly, packaging and testing	Pollution/pollution removal	**Environmental assets** • Freshwater ecosystems • Subterranean freshwater ecosystems • Marine (ocean) ecosystems **Ecosystem services** • Water supply • Water purification • Soil quality regulation	Wastewater is often treated and recycled on site. It contains a range of harmful contaminants (e.g. solvents, arsenic, fine oxide particles) with risks to the environment and human health. Proper treatment is crucial to contain toxic components and recycle chemicals.[37]
EMS & ODM and Hardware			
Raw materials and semiconductors procurement	Refer to semiconductors section for more information on impacts related to raw materials and semiconductors procurement.		

37 Hess, J.C. (2024) Chip Production's Ecological Production: Mapping Climate and Environmental Impact.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
EMS & ODM assembly, testing, packaging Hardware product design and development	Resource use/ replenishment and others Water withdrawal and consumption	**Environmental assets** • Freshwater ecosystems • Water resources **Ecosystem services** • Water supply • Water purification • Water flow regulation	Water-related impacts occur throughout the electronics and hardware manufacturing process as semiconductor fabrication is extremely water intensive. For example, producing a single smartphone requires approximately 3,190 gallons (12,760 litres) of water, much of which represents grey water needed to dilute and treat wastewater generated during manufacturing.[38] These impacts become even more acute during the dry season, particularly in regions with a small number of stressed reservoirs, where freshwater availability is already limited and local environmental boundaries for water use are exceeded. Reduced water supply lowers dilution capacity, exacerbates pollution risks and creates operational vulnerabilities for both industry and local ecosystems.[39]

38 Water Footprint Calculator (2022) The Hidden Water in Everyday Products.

39 Roussilhe, G. et al. (2024) From silicon shield to carbon lock-in? The environmental footprint of electronic components manufacturing in Taiwan (2015–2020).

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
EMS & ODM assembly, testing, packaging Hardware product design and development	Pollution/pollution removal	**Environmental assets** • Land (terrestrial) ecosystems • Freshwater ecosystems • Water resources • Marine (ocean) ecosystems **Ecosystem services** • Provisioning (water supply) • Water purification • Soil and sediment retention • Water flow regulation	Significant quantities of wastewater are generated during the processing of electronic components.[40] Such wastewater results from processes like cleaning, fabrication operations, electroplating and printed circuit board production. They can contain multiple pollutants, including acids and alkalis, heavy metals, organic solvents and fluxes.[41] For example, wastewater arising from printed circuit boards (PCB) production is heavily contaminated with copper, iron and organic matter, which represents a potential pollution source.[42] Water pollution can worsen when factories operate without adequate wastewater treatment systems. Toxic wastewater may impact the environment, including by direct release into rivers through stormwater pipes, through raw sewage overflowing without purification or through leaks from storage tanks.[43]

40 Yadav, N. (n.d.) Electronic Industry Waste Water.

41 ENCORE Partners (2024) ENCORE: Exploring Natural Capital Opportunities, Risks and Exposure.

42 Geric, M. et al. (2017) Environmental risk assessment of wastewaters from printed circuit board production: A multibiomarker approach using human cells.

43 Ellis, B. (2000) Environmental issues in electronics manufacturing: a review.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
E-waste collection, treatment, recycling and disposal	Pollution/pollution removal	**Environmental assets** • Land • Terrestrial ecosystems • Freshwater ecosystems • Water resources • Marine (ocean) ecosystems **Ecosystem services** • Water supply • Water purification • Soil and sediment retention • Soil quality regulation • Global climate regulation • Nursery population and habitat maintenance • Solid waste remediation • Recreation-related services • Visual amenity services	Hardware and electronics production leads to over 60 billion kilogrammes of e-waste produced each year.[44] Global e-waste generation is increasing by around 2.6 billion kg annually and is projected to reach 82 billion kg by 2030.[45] E-waste covers plastics and other waste from electronic devices. Electronic devices can also contain toxic and persistent substances such as mercury and flame retardants. For example, televisions, personal computers, tablets and smartphones contain substantial volumes of plastics.[45] Most flame retardants (80%) are found in screens and monitors. Recycling these products is technically challenging because plastics containing flame retardants need to be identified and separated from other plastics to avoid recirculating hazardous substances. Globally, properly collected and recycled e-waste averages 20-22%,[46] the remaining 80% is undocumented, with much ending up buried underground as landfill.[47] Where collection, treatment and disposal of e-waste are not adequately managed, toxic and persistent substances can leach into soil and freshwater. These releases can contaminate food webs and contribute to habitat degradation and biodiversity loss.

44 WEF (2025) Nature Positive: Role of the Technology Sector.

45 Cordelis, P. et al. (2024) The Global E-waste Monitor 2024.

46 Geneva Environment Network (2025) The Growing Environmental Risks of E-Waste – Geneva Environment Network; Cordelis, P. et al. (2024) The Global E-waste Monitor 2024.

47 Geneva Environment Network (2025) The Growing Environmental Risks of E-Waste – Geneva Environment Network.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
Datacentres (infrastructure)			
Construction of datacentres	Land/freshwater/ ocean use change	**Environmental assets** • Land • Terrestrial (land based) ecosystems **Ecosystem services** • Biomass provisioning • Flood mitigation • Soil quality regulation • Water flow regulation	Building or expanding datacentres can require land clearance and habitat conversion, particularly where development occurs on greenfield or previously undeveloped sites. Further indirect land use change may arise from the construction of access roads, other supporting infrastructure and the increased access to the area this creates for wider human activities. Refer to TNFD Engineering, Construction and Real Estate sector guidance for more details.
Hardware procurement	Land/freshwater/ ocean use change Resource use	**Environmental assets** • Land • Terrestrial (land-based) ecosystems • Mineral and energy resources • Subterranean terrestrial ecosystems **Ecosystem services** • Potentially all	Land-based ecosystems are used for mining natural commodities for IT equipment. This can include land clearing (surface layers including vegetation, water and soils) for open cut mining, underground mining portals, haul ways and waste rock dump deposition. Refer to the TNFD Metals and Mining sector guidance and semiconductors, EMS & ODM and hardware sections for more information on associated upstream impacts.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
Energy sourcing and use	Resource use/ replenishment	**Environmental assets** • Water resources • Freshwater ecosystems **Ecosystem services** • Water supply • Water flow regulation	**Water consumption for electricity generation:** The indirect water consumption of datacentres from power generation sources requires significant water withdrawal and consumption. Refer to TNFD Electric Utilities and Power Generators sector guidance for more details.
Datacentre cooling (with or without water recycling)	Resource use/ replenishment	**Environmental assets** • Water resources • Freshwater ecosystems **Ecosystem services** • Water supply • Water flow regulation	Historically, electric-powered air cooling could meet the requirements of datacentres with server rack power densities of 20 kilowatts or less. As rack densities have increased, direct-to-chip water cooling has become the preferred method, given its higher capability. Liquid cooling methods can be extremely water intensive. On average, a hyperscale facility can use 2.1 million litres of water per day, while a retail facility may use around 68,000 per day. With an average of 60% of water use consumed by evaporation and the remaining 40% going into local wastewater systems, managing water use is a key way to reduce datacentres' impacts on local water supply.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
Datacentre operations	Pollution/pollution removal	**Environmental assets** • Atmospheric systems **Ecosystem services** • Local (micro and meso) climate regulation • Global climate regulation • Air filtration	As datacentres are built to run around the clock, to ensure service continuity during grid outages, the rapid growth of datacentres is accompanied by a surge in on-site backup generator installations.[48] Diesel backup generators used at datacentres are often high-capacity units, ranging in size from 1.5 megawatts to over 3 megawatts each. Larger datacentres may deploy multiple units. Most generators are designed to provide 1.5 to 2 times the total connected load, and many are diesel-powered. These diesel generators emit significant amounts of particulate matter (PM), nitrogen oxides (NOx), sulphur dioxide (SO_2) and carbon dioxide (CO_2) – pollutants that degrade air quality, contribute to climate change and can pose serious health risks to nearby communities.[49]
Datacentre operations	Pollution/pollution removal	**Environmental assets** • Terrestrial (land based) ecosystems • Freshwater ecosystems **Ecosystem services** • Soil quality regulation • Water supply • Water regulation • Water purification	As water circulates within the system of a datacentre, it absorbs heat, resulting in effluent temperatures of 30 to 40°C, while accumulating chemical additives like anticorrosive (e.g. molybdate), biocides and leached metals (Cu, Zn). Blowdown from cooling towers can concentrate total dissolved solids (TDS) to 2,000 ppm. Discharge without proper treatment can cause impacts to downstream environments from thermal pollution and chemical excesses.[50]

48 Ren, S. and Wierman, A. (2025) Mitigating the Public Health Impacts of AI Data Centers.

49 Stax Engineering (2025) The environmental impact of data centers.

50 Chelsey, J. (2025) Data Center Water Treatment Systems: In Theory and in Practice.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
E-waste collection, treatment, recycling and disposal	Pollution/pollution removal	Refer to EMS & ODM and hardware section for more information on e-waste related impacts on nature.	
Datacentre infrastructure decommissioning	Pollution/pollution removal	**Environmental assets** • Terrestrial (land based) ecosystems • Freshwater ecosystems **Ecosystem services** • Soil quality regulation	Refer to TNFD Engineering, Construction and Real Estate sector guidance for more information on the impacts of decommissioning infrastructure.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
Telecommunications (infrastructure)			
Construction/ installation of telecommunications infrastructure	Land/freshwater/ ocean use change	**Environmental assets** • Land • Terrestrial (land-based) ecosystems **Ecosystem services** • Nursery population and habitat maintenance • Recreation-related services • Visual amenity services • Spiritual, artistic and symbolic services	Construction requires clearing land, disrupting habitats and reducing biodiversity. Towers often occupy 50 to 200 square metres per site.[51] Telecommunication towers can present risks to birds and wildlife, particularly during migration, as collisions can cause injuries or fatalities. In addition, tower construction and maintenance can disturb habitats, fragment ecosystems and disrupt wildlife corridors.[52] Towers can also impact the visual and cultural landscape, especially in scenic or densely populated areas, requiring thoughtful planning, design and community engagement to balance infrastructure needs with environmental and aesthetic considerations.[53] The installation and maintenance of submarine cable systems can impact the seafloor's benthic habitats and their placement may disturb sensitive habitats. Refer to TNFD Engineering, Construction and Real Estate sector guidance for more details.

51 IFC (2007) Environmental, Health, and Safety Guidelines for Telecommunications.

52 American Bird Conservancy (n.d.) Bird Collisions and Communications Towers – American Bird Conservancy.

53 IFC (2007) Environmental, Health, and Safety Guidelines for Telecommunications.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
Waste and e-waste collection, treatment, recycling and disposal	Pollution/pollution removal	**Environmental assets** • Land • Terrestrial (land-based) ecosystems • Water resources **Ecosystem services** • Water purification • Soil and sediment retention • Soil quality regulation • Global climate regulation • Nursery population and habitat maintenance • Solid waste remediation	For telecommunication organisations, waste includes material waste from telecommunications infrastructure, including towers, cables and satellites. Other sources of waste can relate to e-waste. Refer to EMS & ODM and hardware section for more information on e-waste related impacts on nature.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
Telecommunications infrastructure decommissioning	Pollution/pollution removal	**Environmental assets** • Land • Terrestrial (land-based) ecosystems • Water resources **Ecosystem services** • Water purification • Soil and sediment retention • Soil quality regulation • Global climate regulation • Nursery population and habitat maintenance • Solid waste remediation	Refer to TNFD Engineering, Construction and Real Estate sector guidance for more information on the impacts of decommissioning infrastructure.

Business activity	Driver of nature change	Environmental assets and ecosystem services affected	Impact on nature (description)
Software & IT services, Internet & Media services and Telecommunications services			
Use of datacentre infrastructure	Refer to datacentre (infrastructure) and/or telecommunications (infrastructure) sections for more information on infrastructure-related impacts.		
Software development, deployment and operation	Resource use/ replenishment	**Environmental assets** • Terrestrial ecosystems • Freshwater ecosystems • Water resources **Ecosystem services** • Water supply • Water purification • Water flow regulation	Software products, digital platforms and services require datacentre capacity and can thereby drive water and energy use, including for cooling in datacentres. Reliance on datacentres varies, influenced by computational intensity, efficiency, architecture, lifecycle decisions, scaling and regional deployment choices. For datacentre-related impacts, refer to datacentres (infrastructure) section.

External factors

External factors with relevance to the technology and communications sector include:

Climate change leads to a higher frequency of extreme weather events. These influence provisioning services such as water supply and regulating and maintenance ecosystem services, such as rainfall pattern regulation, with increased droughts in some regions and increased rainfall in others. Technology and communications organisations will need to assess the influence of climate change on the ecosystem services upon which they depend, such as the impacts of water shortages, drought, storm and flood events and the unintended consequences these may have on the manufacture or infrastructure they depend on to operate. For example, East Asia's 2021 drought resulted in a 20% fall in water reservoir capacity impacting semiconductor supply chains globally. This highlights the sector's dependence on water supply and vulnerability when water shortages occur, severely restricting output.[54]

Other actors within the same landscape or water basin may also affect the land and freshwater resources on which technology and communications organisations depend. Organisations should therefore consider cumulative pressures and relevant land stewardship and water governance arrangements.

Organisations should refer to the TNFD biome guidance for more examples of ecosystem services that may be present in the locations where they are operating.

E3

E3: Dependency and impact measurement

Guiding questions:

What is the scale and scope of our dependencies on nature? What is the severity of our negative impacts on nature? What is the scale and scope of our positive impacts on nature?

In E3, organisations move to measure the scale and scope of these dependencies and impacts (noting measurement quality depends on data availability, which may improve over time in consecutive assessment cycles).

For the quantification of dependencies and impacts in the technology and communications sector, organisations may find it useful to refer to the TNFD sector metrics in Section 3 of this guidance.

Metrics linked to the location of assets and/or activities can be used to quantify impact drivers, changes to the state of nature and changes in the supply of ecosystem services. Such assessment metrics are provided in Annex 1 of the TNFD LEAP guidance. These can be provided as an absolute figure, rate of change or as an intensity/efficiency ratio. Intensity or efficiency ratios can provide useful complementary information but may not accurately represent the scale or severity of an impact. For example, water use effectiveness (WUE)

54 Zang, K. (2024) How Water Scarcity Threatens Taiwan's Semiconductor Industry.

should be considered alongside total water consumption, water source, cooling technology and the condition and water stress of the relevant basin (see also Box 3 on water efficiency and impact metrics for datacentres).

Further guidance on measurements, including the use of baselines and reference conditions, is covered in the TNFD recommendations and LEAP guidance.

Measuring water-related dependencies and impacts

Semiconductor fabrication, EMS & ODM and hardware manufacturing, datacentres and some telecommunications operations depend on reliable supplies of water of sufficient quantity and quality. Relevant measurements to consider include:

- Water withdrawal and consumption from areas of water scarcity, disaggregated by facility, source and basin;
- Water recycled and reused;
- Water replenished, including the volume replenished and verified changes in water availability or ecosystem condition in the relevant basin;
- Wastewater discharge, including volume, temperature, quality and pollutant loads; and
- Exposure of operations to high physical water risk, including limited water availability, flooding and poor quality of water, considering basin-level water availability, water quality, other users' requirements and applicable withdrawal limits.

For semiconductor fabrication and electronics manufacturing, measurements can also distinguish water used for ultrapure water production, cleaning and rinsing, cooling, humidity control and other facility operations. For datacentres, organisations can distinguish water used for server-level cooling, facility-level cooling and heat rejection.

Organisations should estimate water withdrawn and consumed through purchased electricity where this is significant separately from water used at the facility. WRI's Guidance for Calculating Water Use Embedded in Purchased Electricity can support such estimations.

Where facility or supplier data are unavailable, organisations can combine production volumes, technology or process information and location-specific water-use factors, disclosing levels of uncertainty. These estimates should be progressively replaced by primary data for priority locations and suppliers.

Measuring pollution and waste impacts

For pollution, relevant measurements to consider include:

- Pollutant releases to air, water and soil, including wastewater volume, temperature and pollutant loads, and effects beyond the site boundary;
- Hazardous and non-hazardous waste by type and treatment route, including e-waste reuse, repair, refurbishment, recycling, recovery and disposal; and
- Changes in receiving ecosystem health, considering cumulative pollutant loads.

Measuring land, ocean and ecosystem impacts

For fabrication and manufacturing facilities, datacentres and telecommunications infrastructure, relevant measurements to consider include:

- The extent and condition of ecosystems affected during construction, operation, expansion and decommissioning;
- Habitat loss, degradation and fragmentation, particularly from linear infrastructure and cable routes;
- Changes in soil or sediment condition; and
- Changes in habitat connectivity and relevant species populations.

Measuring dependencies on ecosystem services

Dependency measurements combine the extent of reliance on an ecosystem service with information on the condition and resilience of the environmental assets that provide it and the supply of the ecosystem service itself.

Relevant examples of measurements of dependencies include:

- The proportion of semiconductor or hardware production capacity dependent on freshwater of a specified quality;
- The proportion of datacentre capacity dependent on water-based cooling or on electricity generation with significant water dependencies;
- The number, capacity or route length of telecommunications assets dependent on flood and storm mitigation, soil and sediment retention or seabed stability;
- For water supply and water flow regulation, changes in the volume or reliability of water available to semiconductor fabs, electronics manufacturing facilities or datacentres at the relevant basin or site;
- For flood and storm mitigation, changes in the level of protection provided by natural buffers to datacentres, telecommunications infrastructure and other exposed assets, including changes in flood frequency or severity; and
- For soil and sediment retention and seabed stability, changes in erosion, sediment movement or slope/seabed stability along telecommunications infrastructure and cable routes.

Table 24 in the LEAP guidance provides further examples of ecosystem service indicators and metrics. Organisations can consider assessing current conditions and trends in the provision of these ecosystem services. The financial consequences of changes in ecosystem service provision are considered subsequently in the Assess phase.

Measuring impacts associated with digital products and services

Software products, digital platforms, media and telecommunications services create physical resource demands through the datacentres, telecommunications networks and hardware used to provide them. Organisations can allocate energy, water, hardware and other resource use to products, services or workloads using measures such as:

- Compute-hours or processing time;
- Electricity consumption;
- Data storage and transmission volumes;
- Workload or hosting location; and
- The number or type of transactions, users, queries or other service units.

These service-unit measures should only be used to allocate physical resource use and related impacts; they are not measures of impact on the state of nature.

Measurement methods

Potential methods and data sources include in-house company data on impact drivers and resource use, such as water withdrawal and consumption, wastewater discharge, energy use and waste generation. Where more detailed site-level or ecosystem information is needed, organisations can draw on environmental impact assessment data, geographic information systems, satellite and other remote-sensing data, field surveys, sensors, camera traps, acoustic monitoring and locally applicable habitat condition or biodiversity metrics.

Primary data should be prioritised for direct operations and value chain locations associated with the most significant dependencies and impacts. Where primary data are unavailable, organisations may use estimates or proxy data, documenting the sources, assumptions, uncertainty and limitations (see also Box 17 of the Guidance on the identification and assessment of nature-related issues: The LEAP approach).

When life cycle assessment and other value chain estimation methods support measurement, assessment boundaries should reflect the activity being analysed:

- For EMS & ODM and hardware, the types of components, equipment and products manufactured and their production and end-of-life pathways;
- For datacentres and telecommunications infrastructure, construction, operation, maintenance, expansion and decommissioning;
- For software and digital services, development, training where applicable, operational use, data transmission and storage, and changes or optimisation; and
- For upstream materials, components and energy, supplier locations, procurement volumes and production technologies.

Measuring positive impacts

Relevant measurements for the technology and communications sector may include improvements in water availability or quality in affected basins, reductions in pollutant loads entering ecosystems, and restoration or improved connectivity of habitats affected by facilities, linear infrastructure or cable routes.

Refer to the Guidance on the identification and assessment of nature-related issues: The LEAP approach for further guidance.

E4

E4: Impact materiality assessment

Guiding question:

Which of our impacts are material?

As for all components, refer to the Guidance on the identification and assessment of nature-related issues: The LEAP approach.



2.4. Assess nature-related risks and opportunities

This section provides additional guidance to help technology and communications organisations with the Assess phase of the LEAP approach.

A1

A1: Risk and opportunity identification

Guiding question:

What are the corresponding risks and opportunities for our organisation?

In the Assess phase, an organisation identifies the nature-related risks and opportunities associated with the dependencies and impacts previously identified in the Evaluate phase. Tables 7 and 8 provide illustrative examples of nature-related risks and opportunities for the technology and communications sector. Unless otherwise specified, the risks and opportunities laid out in the tables below can apply to any of the sub-sectors within the scope of this guidance.

Table 7: Illustrative nature-related risks in the technology and communications sector

Nature-related dependency/impact	Risk category		Exposure pathway	Primary business effect options	Potential response options
Land conversion	Transition	Market	Land developments require conservation measures as investor and community expectations change.[55]	Costs associated with conservation initiatives.	Rehabilitation, conservation initiatives.
		Policy	Emerging requirements on planning, permitting on land use for construction or expansion of datacentre facilities (in USA, Ireland and UK e.g. England's Biodiversity Net Gain policy).	Restrictions on new developments and obligations towards land use.	Implement land management strategy. On-site monitoring of changes to biodiversity.

55 European Committee of the Regions (n.d.) Ecological compensation to offset environmental impact of new data centre.

Nature-related dependency/impact	Risk category		Exposure pathway	Primary business effect options	Potential response options
Soil and water pollution	Transition	Liability	Long-term persistence and bioaccumulation of PFAS and other chemicals when not managed adequately.[56] Retrospective litigation for toxification of water or soil because of poor pollutant management measures.[57]	Increase in contingent liabilities as regulations become more stringent on PFAS and other chemicals. Costs of litigation and fines. Clean-up costs.	Use of less toxic alternative and substitution materials. Implement stricter pollutant management strategies.
		Technology	Lack of timely investments into treatment technology to meet increased regulatory requirements to manage novel substances of concern.[58]	Increase in costs linked to fines/penalties due to pollution and to new waste treatment technologies.	Invest in new waste treatment technologies.

56 Hess, J.C. (2024) Chip Production's Ecological Production: Mapping Climate and Environmental Impact.

57 Tozzi, C. (2025) 4 Ways to Eliminating Data Center Water Pollution.

58 Hess, J.C. (2024) Chip Production's Ecological Production: Mapping Climate and Environmental Impact.

Nature-related dependency/impact	Risk category		Exposure pathway	Primary business effect options	Potential response options
Waste generation and disposal	Transition	Policy	Stricter waste disposal requirements.[59]	Costs associated with fixing inappropriate waste disposal.	Implement stricter waste management strategies.
			Stricter policies for waste circularity and e-waste management.[60]	Increased costs to transition to circular procurement and improved waste management practices.	Implement material efficiency and circular economy initiatives.
		Reputational	Informal or non-compliant e-waste recycling and landfill disposal.	Reduction in revenue due to reduced brand value.	Implement suppliers' certification programmes.
Non-GHG air pollution	Transition	Policy	Stricter national emission regulations.[61]	Increase in cost of damage payouts (e.g. compensation) caused by noncompliance.	Implement stricter air pollution mitigation strategies.
		Reputational	Association of the company with poor management of polluting substances increases regulatory and stakeholder scrutiny of applications for new sites.[62]	Downward revisions to business growth forecasts due to inability to expand and gain access to new sites.	Implement improved pollution management policies. Invest in rehabilitation, conservation initiatives.

59 Tozzi, C. (2025) 4 Strategies for Eliminating Data Center Water Pollution.

60 Gu, H. (2024) The Hidden Environmental Cost of Data Center Growth – Millions of Tons of E-Waste.

61 Lu, H.H. et al. (2023) Continuous Improvements and Future Challenges of Air Pollution Control at an Advanced Semiconductor Fab.

62 Chow, A.R. (2025) 'We are the Last of the Forgotten:' Inside the Memphis Community Battling Elon Musk's xAI.

Nature-related dependency/impact	Risk category		Exposure pathway	Primary business effect options	Potential response options
Acute or chronic water stress	Physical	Acute and/or chronic	Overabstraction for business processes stressing water resources and freshwater ecosystems.[63]	Cost of disruption to operations due to cooling and humidification disruptions and increase in the cost of water management and control.[64] Decline in value of business assets due to lack of availability of natural resources.	Implement stricter water management policies. Implement water catchment management schemes. Develop water use efficiency strategy. Adopt water efficient technologies (e.g. more efficient datacentre cooling systems).

63 Zhang, K. (2024) How Water Scarcity Threatens Taiwan's Semiconductor Industry.

64 S&P Global (2025) Beneath the surface: Water stress in data centers.

Nature-related dependency/impact	Risk category		Exposure pathway	Primary business effect options	Potential response options
Acute or chronic water stress	Transition	Policy	Stricter water quantity and/or quality regulation.[65]	Increase in costs due to fines for failure to meet regulatory limits on water use or water quality requirements. Increase in taxes leading to increase in expenditure. Cost of operational interruptions due to permit denials linked to failure to meet legally binding targets to reduce water use.	Implement stricter water management policies. Develop water use efficiency strategy.
Acute or chronic water stress	Transition	Market	Conflict with other water users over limited water resources.[66]	Increased costs of recycling or alternative sources of water to ensure supply in the face of increased competition over access.	Contribute to collaborative water catchment management schemes.

65 S&P Global (2025) Beneath the surface: Water stress in data centers.

66 Starr, S. (2025) Water levels across the Great Lakes are falling – just as US data centers move in.

Nature-related dependency/impact	Risk category		Exposure pathway	Primary business effect options	Potential response options
Acute or chronic water stress	Transition	Reputational	Reputational damage caused by increasing water stress due to overabstraction from the catchment.[67]	Cost of operational interruptions, delays or restrictions arising from unresolved concerns or grievances involving Indigenous Peoples, Local Communities and affected stakeholders.	Participate in integrated water basin management, with meaningful stakeholder engagement, and support collaborative action to reduce cumulative pressure on water resources.
Land and water use, and pollutant releases	Transition	Liability	Legal claims, fines or remediation obligations arising from impacts on land, freshwater ecosystems or affected stakeholders.	Legal and remediation costs, operational delays and reputational damage.	Strengthen environmental due diligence, compliance and grievance mechanisms; engage affected stakeholders; prevent and remediate impacts.
Raw materials scarcity/ depletion	Physical	Acute or chronic	Disruption of operations due to reduced availability of raw materials.[68]	Increased costs for procuring raw materials for manufacturing processes.	Use alternative or substitute materials with less negative impact on nature. Invest in material efficiency and circularity.
	Transition	Reputational	Increased stakeholder concerns due to high demand for raw materials, including metals, minerals and rare earth elements used in manufacturing.	Reduction in revenue due to reduced brand value, leading to potential loss of contracts or market access and leads.	Implement suppliers' certification programmes and schemes. Implement responsible sourcing policies.

67 Starr, S. (2025) Water levels across the Great Lakes are falling – just as US data centers move in.

68 Morphy, E. (2024) Two mines in North Carolina are the world's only producer of the quartz necessary for semiconductor manufacturing.

Table 8: Illustrative opportunities in the technology and communications sector

Nature-related dependency/impact	Opportunity type	Illustrative opportunity in the technology and communications sector	Primary business benefit/magnitude indicators
Reduced land use Land restoration	Operational efficiency and resilience	Adopting co-location models and/or sharing tower infrastructure reduces the need for single-tenant sites, lowering overall resource use for construction, operation and maintenance of datacentres and/or telecommunications infrastructure.[69]	Cost reductions by reducing deployment, maintenance and material costs over the asset lifetime.[70]
Land restoration	Operational efficiency and resilience	Creating buffer zones around infrastructure to increase ecosystem condition and mitigate the impacts of natural hazards such as flooding, landslides and others.[71]	Reduced revenue losses and costs associated with infrastructure resilience.
	Business model innovation	Selling biodiversity credits emerging from policy requirements on use of land (e.g. England's Biodiversity Net Gain policy).	New revenue streams from biodiversity credits.
Energy use and associated water and resource use	Resource efficiency	Datacentres can provide grid flexibility through demand response, workload shifting and energy storage, supporting grid stability and greater integration of variable renewable energy. This may reduce reliance on water- and resource-intensive electricity generation.	Reduced electricity costs; revenue from flexibility services; improved operational resilience; and reduced costs from indirect water and resource use associated with electricity supply.
Energy and water use for datacentre cooling	Resource efficiency	Deploying liquid cooling systems to reduce cooling energy demand and, where closed-loop systems are used, operational water consumption, while enabling waste-heat recovery.	Lower energy and water costs, increased computing capacity and potential value from recovered heat.

69 OECD (2025) The Environmental Sustainability of Communication Networks.

70 World Bank Group (2024) Resilient telecommunications infrastructure: A practitioner's guide.

71 Bennett, G. and Mulongoy, K.J. (2006) Review of Experience with Ecological Networks, Corridors and Buffer Zones.

Nature-related dependency/impact	Opportunity type	Illustrative opportunity in the technology and communications sector	Primary business benefit/magnitude indicators
Soil and water pollutants reduction	Operational efficiency and resilience	Use of efficient technology and management practices, including wastewater treatment and reuse, helping to maintain clean water supplies within watersheds.[72]	Reduction in operational costs due to improved readiness and response to regulatory changes.
	Operational efficiency and resilience	Advanced electrochemical wastewater treatment, such as electrocoagulation, of heavy metal and chemical discharges from electronics manufacturing and metal processing.	Reduced costs associated with wastewater management.
	New product or service market	Investment in research and development to find alternatives for hazardous chemicals and gases released in manufacturing.[73]	Increased resilience due to business diversification.
	Operational efficiency and resilience	Enhanced sustainability performance, supporting a positive corporate reputation through reduced resource use and management of pollutants discharged.	Increased revenue from improved brand perception.

72 WEF (2024) Why circular water solutions are key to sustainable data centers.

73 Hess, J.C. (2024) Chip Production's Ecological Production: Mapping Climate and Environmental Impact.

Nature-related dependency/impact	Opportunity type	Illustrative opportunity in the technology and communications sector	Primary business benefit/magnitude indicators
Waste minimised, reused or recycled	Supply chain resilience	Increased use of recycled, reused, refurbished equipment, such as the recycling of cable materials, reintroduces materials into supply chain.[74] Take-back programmes, reverse logistics and collection systems facilitate the recovery of critical materials from end-of-life electronics and other products.[75]	Cost reductions tied to depletion of natural resources (e.g. depleting critical materials). Reduced exposure to raw material and natural resource price volatility. Reduced costs linked to reduced reliance on new materials.
	Business model innovation	Recycling or recovery of materials that can be repurposed by other industries.[76] For example, cable recovery unlocks economic opportunities through new products and business models in the circular economy.[77]	New revenue streams from recovered and/or recycled materials.
Non-GHG air pollutants	Operational efficiency and resilience	Utilising scrubbers. Implementing a comprehensive air pollution control strategy.	Avoided penalties from meeting national regulations.

74 Swinhoe, D. (2022) Re-use, refurb, recycle: Circular economy thinking and data center IT assets.

75 Gaustad, G. et al. (2018) Circular economy strategies for mitigating critical material supply issues.

76 Hess, J.C. (2024) Chip Production's Ecological Production: Mapping Climate and Environmental Impact.

77 UNEP-WCMC (2025) Keeping connected: submarine communications cables and ocean life.

Nature-related dependency/impact	Opportunity type	Illustrative opportunity in the technology and communications sector	Primary business benefit/magnitude indicators
Water conservation	Operational efficiency and resilience	Use of efficient technology and management practices to maintain sustainable water supplies within a watershed.[78]	Reduced operational costs due to improved readiness and response to regulatory changes.
		Investments in nature-based solutions in the watershed e.g. restoration of wetlands, forests and other ecosystems.[79]	Reduced costs of water treatment and risk of water stress via restoration of water resources.
		Use of smart water management technologies to reduce water use e.g. more efficient cooling systems, reduced loss and leaks.[80] This can be particularly relevant in sub-sectors with high water-related dependencies and impacts, such as semiconductor fabs and datacentres.	Operational cost savings associated with improved water management practices and reduced water use.
Raw materials	Supply chain resilience	Recycling provides an in-house source (prompt or manufacturing scrap) or a domestic source (post-consumer scrap) of critical materials, reducing reliance on virgin extraction.[81]	Cost reductions tied to depletion of natural resources (e.g. depleting critical materials). Reduced exposure to raw material and natural resource price volatility.
All	Operational efficiency and resilience	Using emerging technologies to improve monitoring of the state of nature and biodiversity such as: AI and machine learning; networked sensors and Internet of Things; spatial data analysis/modelling software (GPS & GIS); bioacoustics and eco acoustics; satellite remote sensing; camera traps; Unmanned Aerial Vehicles (UAVs)/drones; eDNA; and blockchain and mobile applications and digital platforms.	Reduced operational costs from improved monitoring and traceability systems.

78 BW Water (2025) Net water positive: Can semiconductor fabs get there?

79 Craske, B. (2025) What Role Do NBSs Have In Sustainable Data Centre Strategy?

80 Alert Labs (2025) How Water Leak Detection Systems for Data Centers Minimize Damage & Downtime.

81 Global Electronics Council (2021) State of Sustainability Research: Sustainable Use of Resources.

A2

A2: Adjustment of existing risk mitigation and risk and opportunity management

Guiding questions:

What existing risk mitigation and risk and opportunity management processes and elements are we already applying? How can risk and opportunity management processes and associated elements (e.g. risk taxonomy, risk inventory, risk tolerance criteria) be adapted?

As for all components, refer to the Guidance on the identification and assessment of nature-related issues: The LEAP approach.

A3

A3: Risk and opportunity measurement and prioritisation

Guiding question:

Which risks and opportunities should be prioritised?

As for all components, refer to the Guidance on the identification and assessment of nature-related issues: The LEAP approach.

A4

A4: Risk and opportunity materiality assessment

Guiding question:

Which risks and opportunities are material and therefore should be disclosed in line with the TNFD recommended disclosures?

As for all components, refer to the Guidance on the identification and assessment of nature-related issues: The LEAP approach.



2.5. Prepare to respond and report

This section provides additional guidance to help technology and communications organisations with the Prepare phase of the LEAP approach.

The mission of the Kunming-Montreal Global Biodiversity Framework (GBF) is to "halt and reverse biodiversity loss" by 2030, with a 2050 vision of "a world living in harmony with nature". Delivering the transition implied by the GBF will require significant changes to business practices across all sectors. The TNFD has published guidance on nature in transition plans. Transition planning offers a way to manage an organisation's responses and contributions to this transition in a coherent, structured way. The LEAP Prepare phase provides initial guidance for addressing dependencies, impacts, risks and opportunities, laying the groundwork for a nature transition plan.

P1

P1: Strategy and resource allocation plans

Guiding question:

What risk management, strategy and resource allocation decisions should be made as a result of this analysis?

Technology and communications organisations need to consider the most effective actions based on the dependencies, impacts, risks and opportunities identified in earlier phases of LEAP. This typically involves developing a multi-year action plan, prioritising actions over time as access to data improves and an organisation's strategy is developed with relevant stakeholders across the value chain.

The TNFD advises organisations to consider actions within the context of the Science Based Target Network's (SBTN's) AR3T framework (Figure 6). This includes four types of actions: avoid and reduce negative impacts; restore and regenerate; transformation of business models, products, services, markets and investments; and contribute to needed systemic change inside and outside value chains.

Table 9 presents illustrative examples of actions that technology and communications organisations may want to consider, based on the material dependencies, impacts, risks and opportunities identified in the Locate, Evaluate and Assess phases of LEAP.

The actions are mapped using the TNFD's interpretation of the AR3T framework and are intended to support organisations when planning and sequencing actions across their sphere of control and influence. Actions may vary significantly by business model and across industries within the sector.

Figure 6: SBTN's AR3T framework



Table 9: Illustrative actions for technology and communications sector mapped to AR3T

Dependency/ impact driver	Impacts/risks to be mitigated	Examples of actions/responses	SBTN Action Framework (AR3T)				
			Avoid	Reduce	Restore	Regenerate	Transform
Land/ freshwater/ ocean use change	Reduced access to freshwater, vulnerability to extreme weather events including droughts and floods putting infrastructure and operations at risk.	Invest in watershed-based nature solutions, such as wetland restoration and conservation, to reduce flood and drought risks, strengthen climate resilience and deliver wider nature co-benefits.					
	Extensive land conversion due to expansion and new infrastructure developments driven by demand for AI and/ or technology and communications services.	Prioritise new developments in brownfield areas to avoid net new impact.					
		Adopt/implement co-location models for telecom infrastructure operators to avoid construction of new infrastructure and avoid and reduce associated impacts.					

Dependency/ impact driver	Impacts/risks to be mitigated	Examples of actions/responses	SBTN Action Framework (AR3T)				
			Avoid	Reduce	Restore	Regenerate	Transform
Pollution/ pollution removal	Contaminated wastewater or pollutants discharged to air, soil and/or local water bodies.	Avoid pollution and contaminants (e.g. wastewater, waste heat, solid waste) impacting the environment by using advanced waste control systems and setting zero-waste-to landfill standards (e.g. UL2799).					
		Rehabilitate areas damaged by pollutants and waste by investing in and supporting land and water restoration efforts.					
	Pollution from hazardous substances and process chemicals/gases (including wastewater contaminants and air emissions), leading to impacts on freshwater and ecosystems and increased regulatory/ reputational risks.	Implement substitution and controls to reduce the use and release of hazardous substances, including adopting lower-impact chemicals and gases and applying best available abatement and wastewater treatment systems.					
	Increased e-waste generation, loss of valuable materials and upstream resource pressures from primary material demand.	Design products and equipment for circularity, including durability, modularity, repairability and reuse, supported by take-back programmes and recovery of materials through refurbishment and recycling pathways.					
	Reduction of water withdrawal.	Invest in water recycling to reduce water withdrawal from natural water bodies.					

Dependency/ impact driver	Impacts/risks to be mitigated	Examples of actions/responses	SBTN Action Framework (AR3T)				
			Avoid	Reduce	Restore	Regenerate	Transform
Water use/ replenishment	Significant water volumes used towards cooling of datacentre facilities.	Engage with government, water utilities, communities and other landscape stakeholders to inform datacentre siting or expansion, considering cumulative and future water demands, impacts on other users and ecosystems, and opportunities for shared water-efficient or circular infrastructure.					
		Select and secure energy from non-water intensive power generation (e.g. solar, wind). Implement water use efficiency strategy.[82]					
		Invest in sustainable water basin management plans with other stakeholders in the landscape, including conservation and/or restoration of watersheds.					
		For new developments, avoid siting facilities in water-scarce areas or where cooling requirements would place additional pressure on water resources.					
		For new and existing facilities, incorporate vegetation to reduce cooling and heating needs.					
		Monitor and optimise cooling systems for efficiency and conditions (e.g. using AI to detect leaks within piping systems, climate monitoring to switch to free air cooling when conditions allow, raising operating temperatures).					
		Implement closed loop systems and water reuse practices e.g. 'zero-water' cooling in data centres, targeting Water usage Efficiency (WUE) that is close to zero by eliminating evaporative cooling losses and circulating cooling water in a sealed loop and/ or prioritising non-potable water where feasible.					

82 Ceres (2025) Drained by Data: The Cumulative Impact of Data Centers on Regional Water Stress; Ceres (2026) Water Behind the Watts: The Hidden Risk of Powering Data Centers.

Dependency/ impact driver	Impacts/risks to be mitigated	Examples of actions/responses	SBTN Action Framework (AR3T)				
			Avoid	Reduce	Restore	Regenerate	Transform
Resource use/ replenishment	Reliance on raw commodities, including metals and mining, puts natural resources at risk of depletion creating risks for the organisation.	Utilise emerging technologies for improved supply chain transparency and transformation.[83]					
		Optimise hardware and process design with circularity in mind, e.g. utilise recycled materials, substitute high-risk commodities with lower impact alternatives, and design with a modular focus to support repairability.					
	Supply disruption risks from impacts of upstream extraction and processing of materials (including metals and minerals), with potential exposure to regulatory, reputational and operational risks.	Engage suppliers and other value chain partners on traceability, responsible sourcing and supplier performance, supported by procurement policies, supplier due diligence and monitoring.					
	High water use, increasing pressure on local water resources and exposure to water scarcity.	Monitor and optimise cooling systems for efficiency and local conditions, including liquid cooling and closed-loop systems where these reduce water use.					

83 Trade-offs should be carefully assessed to ensure the benefits outweigh potential adverse impacts of emerging technologies.

Dependency/ impact driver	Impacts/risks to be mitigated	Examples of actions/responses	SBTN Action Framework (AR3T)				
			Avoid	Reduce	Restore	Regenerate	Transform
All	Impacts associated with inefficient architectures, over-provisioning, excessive data movement or compute-intensive design patterns.	Develop product roadmaps, architecture standards, engineering practices and digital governance processes.					

For additional illustrative examples of corporate responses to mitigate and manage nature-related dependencies, impacts, risks and opportunities (DIROs) in the technology and communications sector, organisations may refer to the World Economic Forum report Nature Positive: Role of the Technology Sector (2025). The report includes examples of actions focused on semiconductors, hardware and electronics, and datacentres. Figure 7 provides a categorisation of selected actions, which WEF has categorised as 'foundational', 'leading' and 'aspirational' actions.

While the scope of the WEF report is narrower than this TNFD technology and communications sector guidance (and does not cover all sector business models), the examples can be relevant for many organisations in the sector, including where material dependencies and impacts arise across value chains.

Large technology and communications organisations that do not have direct operations where the most significant nature-related dependencies and impacts occur (including those in software and IT services, internet media and services and telecommunication services) can engage with suppliers and other value chain partners, via supplier engagement models, contractual levers and collaborative initiatives, to support and encourage implementation of such actions beyond their own direct operations.

Figure 7: Tiering actions based on nature leadership

Foundational	Leading	Aspirational
Foundational, often incremental actions that are, or are becoming, common practice	Actions adopted by several leaders that expand nature benefits and competitive advantage	Ambitious, often transformative and less common practices that shape the value chain
1.1 Review sites for water stress	1.3 Assess complete water footprint	1.5 Restore local watersheds
1.2 Design & operate for efficiency	1.4 Closed-loop & water reuse	2.3 Embed digital circularity practices
2.2 Extend equipment lifespan	2.1 Design for circularity/repairability	2.4 Develop repair infrastructure
2.7 Adopt pollution & waste controls	2.5 Establish collection programmes	2.6 E-waste recycling infrastructure
3.1 Monitor for direct GHG leaks	3.2 Utilise gas scrubbers	2.8 Invest in pollution rehabilitation
4.6 Engage community stakeholders	3.4 Invest in carbon credits	3.3 Design to lower embodied carbon
5.1 Low-carbon onsite energy or PPAs	4.1 Prioritise brownfield development	4.2 Assess biodiversity risk in sites
5.3 Design power-efficient buildings	4.4 Green roofs, native vegetation	4.3 Establish biodiversity baseline
6.1 Suppliers with sustainability certificates	5.2 Sponsor low-carbon capacity	4.5 Invest in biodiversity offsets
7.2 Report nature impacts	5.4 Optimise cooling systems	6.2 Lower-impact metals, minerals
	5.5 Dynamic: process management	6.5 R&D low impact chemicals, gasses
	6.3 Lower-impact chemicals, gasses	7.3 Sector-level nature benchmarks
	6.4 Lower-carbon building materials	
	6.6 Set supply chain commitments	
	7.1 Proactively engage policy-makers	

■ Water use ■ Pollution and waste ■ Greenhouse gas emissions ■ Land use ■ Electricity use ■ Supply chain engagement ■ Policy and external engagement

Source: WEF (2025) Nature Positive: The role of the technology sector.

P2

P2: Target setting and performance management

Guiding question:

How will we set targets and define and measure progress?

Technology and communications organisations may find it useful to consider the following resources to set targets:

- SBTN methods for setting science-based targets related to freshwater, land, ocean and how these methods support biodiversity. Organisations referring to SBTN resources should always consult the latest available guidance and tools on the SBTN website;
- The CEO Water Mandate guidance on setting effective site water targets informed by catchment context (contextual water targets);
- The International Water Stewardship Standard Version 2.0 for setting water-related targets; and
- IUCN's Rapid High-Integrity Nature-positive Outcomes (RHINO) approach, which helps organisations guide interventions to deliver nature-positive outcomes by determining actions to reduce species extinction risk, specifically for vertebrate species.

Illustrative examples of setting water-related, science-based targets for the sector include:

- Setting a time-bound goal or target for the protection or restoration of ecosystems critical to freshwater and aquatic biodiversity; and
- Setting a time bound risk-differentiated, science-based or contextual water use/ replenishment target to address value chain impacts on water availability.

P3

P3: Reporting

Guiding question:

What will we disclose in line with the TNFD recommended disclosures?

As for all components, refer to the Guidance on the identification and assessment of nature-related issues: The LEAP approach.

Additionally, when considering international interoperability throughout their disclosure process, organisations may find it helpful to refer to the following documents:

- TNFD – European Financial Reporting Advisory Group (EFRAG) Correspondence Mapping;
- TNFD – Global Reporting Initiative (GRI) Standards Interoperability Mapping;
- Guidance on Integrating nature-related aspects in Business Responsibility & Sustainability Reporting (BRSR) disclosures using the TNFD recommendations for organisations in India; and
- TNFD-CDP Correspondence Mapping.

P4

P4: Presentation

Guiding question:

Where and how do we present our nature-related risk disclosures?

As for all components, refer to the Guidance on the identification and assessment of nature-related issues: The LEAP approach.

List of datasets and tools

A list of datasets and tools relevant to the technology and communications sector is provided in the reference list.

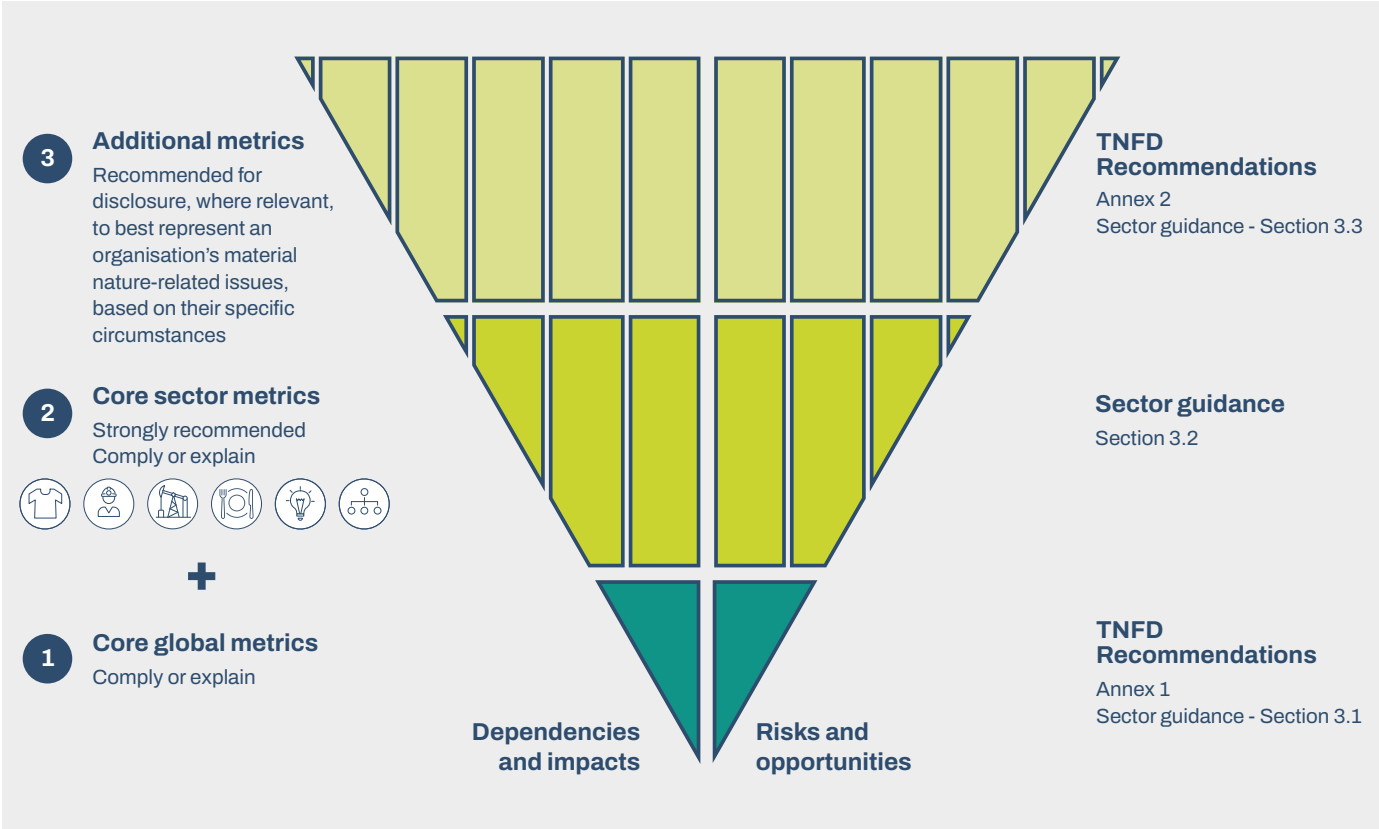
3. Sector-specific disclosure metrics and related guidance

Sector-specific metrics form an important part of the TNFD's measurement architecture (see Figure 8). This reflects the diversity of business models across value chains and their interface with nature across and within sectors. Sector-specific metrics help financial institutions to compare organisations within the same sector, which often face similar nature-related issues.

This section provides the TNFD's sector-specific metrics for the technology and communications sector. It includes:

- Guidance on the application of the core global disclosure indicators and metrics to the technology and communications sector (Section 3.1); and
- Core and additional disclosure indicators and metrics for the technology and communications sector (Sections 3.2 and 3.3).

Figure 8: TNFD disclosure metrics architecture signposted to metrics lists



Where available, the TNFD's recommended metrics for disclosure draw from a range of existing standards and frameworks including the IFRS Sustainability Disclosure Standards, Sustainability Accounting Standards Board (SASB) Standards, GRI Standards, the CDP disclosure platform, the Kunming-Montreal Global Biodiversity Framework and other relevant UN frameworks, the European Sustainability Reporting Standards (ESRS) and others.

Several organisations, including standard-setting organisations, continue to work on identifying relevant sector-level assessment and reporting metrics. The Taskforce recommends that report preparers stay engaged with year-on-year progress on these developments and implement the latest definitions within their risk management processes and disclosures.

Organisations in the technology and communications sector should refer to Annex 1 of the TNFD recommendations for further information on the core global disclosure metrics, which should be reported on a comply or explain basis, except for the placeholder metrics. Where organisations are unable to report against any of the core global metrics, they should provide a short explanatory statement as to why they have not reported those metrics. An organisation should report on the core global disclosure metrics unless:

- It has not been identified as relevant and material to the organisation, e.g. not relevant to business activities or the location the organisation is operating in, or not found to be a material issue for the organisation; or
- It has been identified as relevant and material, but the organisation is unable to measure it due to limitations with methodologies, access to data or because the information is commercially sensitive. In this case, organisations should explain how they plan to address this in future reporting periods.

An organisation should report on the same basis for the core sector disclosure metrics outlined in Section 3.2.

Organisations are also encouraged to draw on the TNFD additional sector disclosure indicators and metrics outlined in Section 3.3 and any other relevant metrics to represent most accurately the organisation's nature-related dependencies, impacts, risks and opportunities.

Box 2: Key considerations for value chain disclosures

For many technology and communications organisations, material nature-related dependencies and impacts will likely arise outside direct operational control, including through leased, co-located or third-party infrastructure.

In this sector, large software, media and telecom companies depend on datacentre infrastructure, whether owned, leased or otherwise contracted. Such datacentres in turn rely on semiconductors and other electronic components. Organisations should assess and disclose material nature-related dependencies and impacts wherever they occur in the value chain, rather than limiting disclosure to activities under direct operational control.

To note, organisations in many other sectors of the economy, beyond the technology and communications sector that is the scope of this guidance, also use datacentres. These nature-related issues will also be relevant to their nature-related assessments and disclosures.

A hypothetical illustrative example

A technology company uses 10% of the capacity of each of 10 datacentres. Two of these datacentres are located in water-scarce areas, where water withdrawal has been identified as material because it adds pressure to local freshwater ecosystems and species dependent on these ecosystems, and may affect other water users (including local communities and farmers), which leads to physical and transition risks for the company.

The company discloses the TNFD metrics following this sector-specific guidance, including core global disclosure metric C3.0 for each of the two datacentres in water-scarce locations under Metrics and Targets B, together with the location and contextual information under Strategy A and Strategy D (see TNFD recommendations).

Each datacentre withdraws 100,000 m^3/year of water, and capacity use is viewed as a reasonable basis for allocation, so the company attributes 10% of the total withdrawal to its own use, i.e. 10,000 m^3/year at each site. The disclosure shows the total withdrawal of water at each site, the amount attributed to the company and the allocation method used.

Improving value chain disclosures over time

Where primary data are not available or comprehensive, for instance, for material impacts upstream or downstream in the value chain, proxy data can be used (see Box 5 of the TNFD recommendations). Organisations should explain the basis of the metric, key assumptions and allocation methods.

Organisations should work with value chain partners to improve data availability and quality over time, including through supplier engagement, contractual levers and industry collaboration. The accuracy and granularity of value chain metrics can thereby improve over subsequent reporting cycles, with transparency on data sources, assumptions and limitations in each reporting period.

3.1. Guidance on the application of the core global disclosure metrics

This section provides guidance, where relevant, on how to apply the TNFD core global disclosure metrics in the technology and communications sector. If no further sector-specific guidance is provided, organisations should refer to the core global disclosure metrics in the TNFD recommendations. As outlined above, core global disclosure metrics should be reported on a comply or explain basis, following the guidance for the technology and communications sector, where provided.

The TNFD encourages organisations to consider and report against placeholder indicators on invasive alien species and the state of nature, where possible, but these are not expected on a comply or explain basis. There are not yet widely accepted metrics for these indicators, but the Taskforce recognises their importance and will continue to work with knowledge partners to develop further guidance on these metrics.

In line with existing TNFD guidance, the following disclosure guidance is intended for a) direct operations of industries mentioned in bold under the column 'Guidance for this sector' and b) for industries with business operations that have material nature-related issues via other industries in their value chains. Industries under b) should specify disclosures on the values (e.g. km^2, m^3) impacted or depended on by their own direct operations and the values impacted or depended on through their value chain. For reference, see the sections on value chain reporting on pages 66 to 67 in the TNFD's recommendations and Box 5 on the use of secondary data in the same document.

In Table 10, 'no further sector specific guidance' does not mean no disclosure is required. In such cases, organisations are referred to the existing disclosure metrics guidance in Annex 1 of the TNFD recommendations.

Table 10: Guidance on the application of the core global disclosure metrics

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
Driver of nature change: Land/freshwater/ocean-use change				
C1.0	Total spatial footprint	Total spatial footprint (km^2) (sum of): • Total surface area controlled/ managed by the organisation, where the organisation has control (km^2); • Total disturbed area (km^2); and • Total rehabilitated/ restored area (km^2).	**Semiconductors, EMS & ODM and Hardware** No further sector specific guidance; refer to the core global disclosure metric. **Datacentres** In reporting this core global disclosure metric, an organisation should include the total surface area (km^2) controlled or managed by the organisation. This should cover the footprint of the physical building(s), as well as land for associated operations, such as outdoor power and cooling equipment, modules, car parks and other operational infrastructure. Any land cleared of vegetation for drainage, setback or buffer requirements should also be included. For areas developed in phases, such as datacentre campuses, the total surface area controlled or managed by the organisation should include all land under operational control, whether reserved for future development or not (e.g. including land under control, but not developed in the reporting period). Where energy infrastructure is constructed primarily to serve a datacentre (e.g. dedicated infrastructure), an organisation should include the associated land footprint in the total spatial footprint of the datacentre. Where such infrastructure is shared or serves multiple purposes, organisations should disclose the total land area for the infrastructure and provide additional context on their use of the energy infrastructure in relation to its total capacity.	TNFD

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C1.0 (cont.)			In disclosing the total rehabilitated or restored area, a datacentre operator should include both surface area controlled or managed by the organisation and areas in the wider landscape/beyond operational boundaries (km^2). **Telecommunication Infrastructure** Refer to TNFD's Additional sector guidance on engineering, construction and real estate. **Telecommunications Services, Software & IT Services, Internet Media & Services** No further sector specific guidance; refer to the core global disclosure metric.	

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C1.1	Extent of land/ freshwater/ ocean use change	Extent of land/ freshwater/ocean ecosystem use change (km^2) by: • Type of ecosystem;[84] and • Type of business activity.	**Semiconductors, EMS & ODM, Hardware, Software & IT Services, Internet Media & Services, Telecommunications Services** No further sector specific guidance; refer to the core global disclosure metric. For upstream operations, an organisation should refer to the TNFD additional sector guidance on Metals and Mining. **Datacentres, Telecommunications Infrastructure** In reporting this core global disclosure metric, an organisation should include land-use change resulting from the construction of new facilities and the expansion of existing facilities for operational purposes. For telecommunications infrastructure, an organisation should also disclose: • Temporary land-use change and temporary disturbances (ha) during telecom construction (e.g. subsea cables installation); and • Permanent land-use change (ha) (e.g. telecommunication deployment, new access roads for permanent deployment). For datacentres and telecommunications infrastructure, an organisation should refer to TNFD's Additional sector guidance on Engineering, Construction and Real Estate.	TNFD

84 When disclosing on extent of ecosystem types users should refer to the International Union for Conservation of Nature (IUCN) Global Ecosystem Typology.

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C1.1	Extent of land/ freshwater/ ocean use change	Extent of land/ freshwater/ ocean ecosystem conserved or restored (km^2), split into: • Voluntary; and • Required by statutes or regulations.	**All Technology sub-sectors** No further sector specific guidance; refer to the core global disclosure metric.	
C1.1		Extent of land/ freshwater/ ocean ecosystem sustainably managed[85](km^2) by: • Type of ecosystem;[86] and • Type of business activity.	**All Technology sub-sectors** No further sector specific guidance; refer to the core global disclosure metric.	

85 Refer to TNFD Glossary.

86 When disclosing on extent of ecosystem types users should refer to the International Union for Conservation of Nature (IUCN) Global Ecosystem Typology.

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
Driver of nature change: Pollution/pollution removal				
C2.0	Pollutants released to soil split by type	Pollutants released to soil (tonnes) by type, referring to sector specific guidance on types of pollutants.	**Semiconductors, EMS & ODM, Hardware and Datacentres** An organisation should disclose individual substances of concern released to soil (in tonnes) and their pollutant class, including but not limited to substances in the following pollutant classes: • Per and polyfluoroalkyl substances (PFAS); • Persistent organic pollutants (POPs); • Heavy metals (e.g. cadmium, lead, mercury); and • Chemical additives (e.g. biocides, corrosion inhibitors). Disclosures should include releases to soil from e-waste treatment, recycling and disposal. An organisation should also refer to applicable legal or regulatory frameworks for pollutants that may be reported as listed in Annex 4. **Telecommunications Infrastructure and Services, Software & IT Services, Internet Media & Services** No further sector specific guidance; refer to the core global disclosure metric.	TNFD

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C2.1	Wastewater discharged	Volume of water discharged (m^3), split into: • Total; • Freshwater; and • Other.[87] Including: • Concentrations of key pollutants in the wastewater discharged, by type of pollutant, referring to sector specific guidance for types of pollutants; and • Temperature of water discharged, where relevant.	In reporting this core global metric, an organisation should include: • pH (acidity/alkalinity); and • pollutant classes[88] listed in Annex 3, which provides illustrative, non-exhaustive examples of substances of concern within each class. Quantitative disclosures on concentrations should be an average over the reporting period, including disclosure of the method used to calculate the average. Organisations should also refer to applicable legal or regulatory frameworks in Annex 4, and disclose any additional information on wastewater discharge, if such frameworks impose requirements not included in this guidance. **Semiconductors, EMS & ODM and Hardware, Datacentres** In reporting this core global disclosure metric, an organisation should disclose: • Total volume of water discharged by destination (m^3): surface water; groundwater; seawater and third parties e.g. municipal sewerage systems or other users;[89] • Total volume of water discharged, split by treatment route (m^3): treated on site before discharge; sent to third-party treatment and discharged without treatment; and • Total volume of water discharged to areas facing high physical water risk (m^3)[90] split into total, freshwater and other water. Concentrations of pollutants in the wastewater discharged by destination (see Annex 3). Reported concentrations should represent an average over the reporting period. Organisations should disclose the method used to calculate these concentrations.	**Semiconductors:** TNFD GRI 303 (2018) Sandhu, S. et al. (2025) **Datacentres:** TNFD GRI 303 (2018) Tozzi, C. (2025) APLD (n.d.) CDC (2025) SA Water (2021)

87 Freshwater: (≤1,000 mg/L Total Dissolved Solids). Other: (>1,000 mg/L Total Dissolved Solids). Reference: GRI (2018) GRI 303-4 Water discharge.

88 For example, where PFAS are present in wastewater discharge, an organisation should disclose the concentration of the specific substances, where known, rather than reporting only the pollutant class.

89 For definitions of surface water; groundwater; seawater and third-party water refer to GRI (2018) Water and Effluents Glossary.

90 For the definition of physical water risk refer to the TNFD Glossary.

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C2.1			Where datacentre facilities use cooling and/or humidification systems that require water, an organisation should disclose: • Volume of water discharged (m^3) at temperatures greater than 30°C, or 2-3°C above ambient levels when reaching natural water bodies; • Volume of water discharged (m^3) to areas facing high physical water risk,[91] split into treated and untreated water, including a description of treatment methods; and • Volume of cooling effluent or other proprietary chemical blends discharged (including waterless and non-conducive blends) (m^3). **Telecommunications Infrastructure and Services, Software & IT Services, Internet Media & Services** No further sector specific guidance; refer to the core global disclosure metric.	

91 For physical water risk refer to WRI Aqueduct Water Risk Atlas and WWF Water Risk Filter.

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C2.2	Waste generation and disposal	Weight of hazardous and non-hazardous waste generated by type (tonnes), referring to sector-specific guidance for types of waste. Weight of hazardous and non-hazardous waste (tonnes) disposed of, split into: • Waste incinerated (with and without energy recovery); • Waste sent to landfill; and • Other disposal methods.	**Semiconductors, Telecommunication Services, Software & IT Services, Internet & Media Services** No sector-specific guidance; refer to the core global disclosure metric on this page and the next. **EMS & ODM and Hardware** In reporting this core global disclosure metric for own operations, refer to the core global disclosure metric. For EEE products placed on the market, an organisation should disclose: • Weight of EEE products placed on the market (tonnes); • Weight (tonnes) and proportion (%) of EEE products placed on the market that is covered by certified take-back, recovery or producer responsibility programmes and designed, labelled or verified for disassembly/repair and component recovery; and • Weight (tonnes) and proportion (%) of e-waste and battery waste collected from EEE products through recovery schemes and/or via take-back programmes, and proportion (%) that is reused, recycled, subject to other recovery operations, and diverted from landfill. **Datacentres and Telecommunications Infrastructure** In reporting this core global disclosure metric for EEE and batteries used by the organisation itself, an organisation should disclose: • Weight (tonnes) of EEE and batteries reaching end of use by the organisation, including from IT upgrade cycles, refurbishment activities, and the proportion (%) that is reused, recycled, or other recovery operations;	**Semiconductors** TNFD **Electronics and Hardware** TNFD Open Compute Project (2024)

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C2.2 (cont.)		Weight of hazardous and non-hazardous waste (tonnes) diverted from landfill, split into waste: • Reused; • Recycled; and • Other recovery operations.	• Weight of e-waste and battery waste generated (tonnes) from IT upgrade cycles, refurbishment activities and battery replacement programmes; and the proportion (%) diverted from landfill and processed by authorised or certified recyclers (downstream e-waste); and • Quantity (tonnes) and proportion (%) of certified e-waste management (recycle, reuse, recovery, etc) by third parties, specifying the certification scheme (e.g., e-Stewards, R2v3).[92] For a datacentre facility, where power generation for the facility is under direct operational control, an organisation should disclose waste from power generation in line with the TNFD sector guidance Electric Utilities and Power Generators. For further clarification on e-waste and battery waste, refer to the TNFD Glossary.	

92 Disclosure of certification coverage does not imply full alignment with the TNFD recommendations and does not replace other TNFD disclosure guidance.

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C2.3	Plastic pollution	Plastic footprint as measured by total weight (tonnes) of plastics (polymers, durable goods and packaging) used or sold broken down into the raw material content. For plastic packaging, percentage of plastics that is: • Reusable; • Compostable; • Technically recyclable; and • Recyclable in practice and at scale.	**EMS & ODM and Hardware** In reporting this core global disclosure metric, an organisation should disclose the weight of plastics (tonnes) incorporated in the final product. **Semiconductors, Datacentres, Telecommunications Infrastructure and Services, Software & IT Services, Internet Media & Services** No further sector specific guidance; refer to the core global disclosure metric.	TNFD Cordelis, P. et al. (2024) EnviroPass (n.d.) A&C Plastics (n.d)

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C2.4	Non-GHG air pollutants	Non-GHG air pollutants (tonnes) by type: • Particulate matter ($PM_{2.5}$ and/or PM_{10}); • Nitrogen oxides (NO_2, NO and NO_3); • Volatile organic; compounds; • (VOC or NMVOC); • Sulphur oxides (SO_2, SO, SO_3, SO_x); and • Ammonia (NH_3).	**Semiconductors** In reporting this core global disclosure metric, an organisation should disclose the following non-GHG air pollutants (tonnes): • Acidic gases (e.g. H_2SO_4, HF, HCl, HNO_3 and H_3PO_4); • Alkaline gases (e.g. NH_3); and • Hazardous air pollutants/toxic gases: PH_3, AsH_3, B_2H_6, Cl_2, O_3, $SiCl_2H_2$, WF_6, SiH_4. **Datacentres** In reporting this core global disclosure metric, an organisation should disclose the following non-GHG air pollutants (tonnes): • Carbon monoxide (CO); and • Hazardous Air Pollutants (HAPs). Where the datacentre operational boundary includes power generation, an organisation should disclose non-GHG air pollutants from power generation in line with the TNFD sector guidance for Electric Utilities and Power Generators. **EMS & ODM, Hardware, Telecommunications Infrastructure & Services, Software & IT Services, Internet Media & Services** No further sector specific guidance; refer to the core global disclosure metric.	**Semiconductors:** EPA (2025) Nguyen, M.T.N. and Lee, J.S. (2024) Yin, Y. and Yang, Y. (2025) **Datacentres:** GRI 305 (2016) EPA (2025) EU Industrial Emissions Directive (2010)

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
Driver of nature change: Resource use/replenishment				
C3.0	Water withdrawal and consumption from areas of water scarcity	Water withdrawal and consumption[93] (m^3) from areas of water scarcity, including identification of water source.	**Semiconductors** In reporting this core global disclosure metric, an organisation should report in areas of water scarcity: • Water withdrawal and consumption (m^3) by use: process (production, including for ultrapure water/UPW), scrubber, cooling and other (specified); • Volume (m^3) of UPW consumption during manufacturing; and • Water withdrawal and consumption (m^3) per unit produced. **Datacentres** For all datacentre facilities, regardless of cooling system and design, an organisation should disclose for areas of water scarcity: • Water consumption (m^3) for electricity generation. Refer to the TNFD Electric Utilities and Power Generators sector guidance[94] for further guidance[95]. For datacentre facilities using cooling and/or humidification systems that require water, an organisation should disclose in areas of water scarcity: • Water withdrawal and consumption in datacentre operations (m3) split by water used for cooling, humidification and other (specifying cooling and humidification technologies and 'other uses'); • Water usage effectiveness (WUE):[96] ratio of the water usage for the datacentre operations (m^3), divided by the energy consumption of the IT computing equipment (kWh) over the reporting period (see Box 3); and • Cycles of concentration: number of times water circulates through a recirculating evaporative cooling system before discharge becomes necessary. The calculation compares dissolved solids concentration in circulating water to makeup water concentration. An organisation should disclose the methodology used.	**Semiconductors:** TNFD Hess, J.C. (2024) Jones, C. and Vacuum, E. (2022) **Datacentres:** TNFD GRI 303 (2018) Genesis Water Tech (2025) Lei, N. et al. (2025) ISSB metric TC-ES-140a.1 for EMS & ODM

93 Water consumption is equal to water withdrawal less water discharge. Reference: GRI (2018) GRI 303-5.

94 Refer to metric C3.0. in the TNFD Sector Guidance for Electric Utilities and Power Generators (p64).

95 Where full quantification is not feasible an organisation should disclose energy mix (by source and geography) or use modelled estimates or industry-average water and energy intensity factors.

96 See Box 3 below on water usage effectiveness, including the disclosure of the definition, boundary and methodology used.

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C3.0 (cont.)			**Semiconductors, EMS & ODM, Hardware, Datacentres** • Total water withdrawn and total water consumed (m^3) from all locations (including but not limited to areas of water scarcity); and • Volume (m^3) and proportion (%) of the total water withdrawn and total water consumed that is from areas of water scarcity. **Telecommunications Infrastructure and Services, Software & IT Services, Internet Media & Services** No further sector specific guidance; refer to the core global disclosure metric.	

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C3.1	Quantity of high-risk natural commodities sourced from land/ocean/ freshwater	Quantity of high-risk natural commodities[97] (tonnes) sourced from land/ocean/ freshwater, split into types, including proportion of total natural commodities.	**Semiconductors, EMS & ODM and Hardware** In reporting this core global disclosure metric, an organisation should disclose the quantity (tonnes) of high-risk natural commodities split into types, used in manufacturing and assembly operations, and the proportion (%) traceable to country of origin, to sub-national level origin (e.g. state/province/ jurisdiction) and, where available, to the extraction site or mine. In addition to the high-risk natural commodities included in the SBTN High Impact Commodity List[13], an organisation should disclose chromium, cobalt, high purity quartz, silica sand, manganese, natural graphite, palladium, rare earth elements, tantalum, tin, titanium and tungsten. An organisation should also consider any other natural commodities at high risk of nature-related impacts during extraction or processing, including those identified through local, regional or national critical-material screening lists, such as the EU Critical Raw Materials list. **Datacentres, Telecommunications Infrastructure & Services, Software & IT Services, Internet Media & Services** No sector-specific guidance; refer to the core global disclosure metric.	**Electronics and hardware:** TNFD Althaf, S. and Babbitt, C.W. (2020) The EU Critical Raw Materials List

97 Users should refer to the Science Based Targets Network (SBTN) High Impact Commodity List (HICL) and indicate what proportion of these commodities represent threatened and CITES listed species.

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C3.1 (cont.)		Quantity of high-risk natural commodities[98] (tonnes) sourced under a sustainable management plan or certification programme, including proportion of total high-risk natural commodities.	**Semiconductors, EMS & ODM and Hardware** In reporting this core global disclosure metric, an organisation should disclose the quantity (tonnes) and proportion (%) of high-risk natural commodities sourced that are covered by recognised responsible sourcing, sustainable management or certification programmes, broken down by commodity and programme. **Datacentres, Telecommunications Infrastructure and Services, Software & IT Services, Internet Media & Services** In reporting this core global disclosure metric, an organisation should disclose the proportion (%) of procured electronic and electrical equipment (EEE) covered by recognised responsible sourcing or certification programmes, broken down by product and programme. Examples of recognised responsible sourcing and certification programmes include the Initiative for Responsible Mining Assurance (IRMA), Aluminium Stewardship Initiative (ASI), The Copper Mark, the Responsible Minerals Assurance Process (RMAP), Fairtrade Gold, Fairmined.[99]	TNFD

98 Users should refer to the Science Based Targets Network (SBTN) High Impact Commodity List (HICL) and indicate what proportion of these commodities represent threatened and CITES listed species.

99 Disclosure of certification coverage does not imply full alignment with the TNFD recommendations and does not replace other TNFD disclosure guidance.

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C4.0	Placeholder indicator: Measures against unintentional introduction of invasive alien species (IAS)	Proportion of high-risk activities operated under appropriate measures to prevent unintentional introduction of IAS, or low risk designed activities.	No further sector specific guidance; refer to the core global disclosure metric.	

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C5.0	Placeholder indicator: Ecosystem condition	For those organisations that choose to report on state of nature metrics, the TNFD encourages them to report the following indicators and to refer to the TNFD additional guidance on measurement of the state of nature in Annex 2 of the LEAP approach: • Level of ecosystem condition by type of ecosystem and business activity; and • Species extinction risk.	No further sector specific guidance; refer to the core global disclosure metric.	

Metric no.	Core global indicator	Core global metric	Guidance for this sector	Source
C5.0	Placeholder indicator: Species extinction risk	There are a number of different measurement options for these indicators. The TNFD continues to work with knowledge partners to increase alignment.	No further sector specific guidance; refer to the core global disclosure metric.	

Box 3: Water efficiency and impact metrics for datacentres

Datacentre water use can vary significantly by cooling technology, operating conditions and location. Facility-level metrics are therefore important for understanding both operational water efficiency and pressure on local water resources.

Water usage effectiveness (WUE)[100] is established in the datacentre sector and relates annual site water use to IT equipment energy consumption. WUE is also standardised under ISO/IEC 30134-9.[101] However, reported values may still differ depending on the measurement boundary and calculation approach used. In line with the TNFD recommendations, organisations should disclose WUE by location and explain the definition, boundary and methodology used. This supports interpretation, comparability and further market convergence.

Water usage impact (WUI)[102] complements WUE by combining site water consumption with local water stress. WUI is a recent metric and is not currently widely used for benchmarking across datacentres; its methodology may evolve as adoption increases. It is therefore included as a TNFD additional sector metric. Organisations reporting WUI should disclose the metric version, water-consumption definition, water-stress source and reference year used.

3.2. Core sector disclosure indicators and metrics

The TNFD core sector disclosure metrics for the technology and communications sector are outlined in Table 11. These metrics are recommended by the TNFD to be disclosed by all report preparers in this sector on a comply or explain basis, in addition to reporting the core global metrics outlined above. Some of these metrics are applicable for a limited number of sub-sectors, which have been highlighted where relevant. All other metrics apply to all sub-sectors within the scope of this guidance.

100 The Green Grid (2011) WP #35 – Water Usage Effectiveness (WUE): A Green Grid Data Center Sustainability Metric.

101 ISO (2012) ISO/IEC 30134-9: Information technology – Data centres key performance indicators – Part 9: Water Use Effectiveness (WUE).

102 The Green Grid (2025) WP#95 Water Usage Impact (WUI) v1 Metric.

Table 11: Core sector disclosure indicators and metrics

Metric no.	Metric category	Metric subcategory	Core sector disclosure indicator	Core sector disclosure metric	Source
T.C1.0	Impact driver	Land/ freshwater/ ocean use change	Habitat fragmentation	**Telecommunications infrastructure** Length (km) of new or expanded linear telecommunications infrastructure constructed outside existing infrastructure corridors (including terrestrial and subsea cables, fibre-optic routes, ducts/conduits and overhead telecommunications lines): • In sensitive locations, by sensitive location criteria met, stating the ecosystem type; and • In other areas, stating the ecosystem type.	Adapted from TNFD Engineering, Construction and Real Estate guidance
T.C2.0	Impact driver	Pollution/pollution removal	Pollution incidents to air, soil and/or water	Number of pollution incidents involving releases to air, soil and/or water that required notification to a competent authority, required remediation or containment, or exceeded permit, licence or legal limits including location, substance spilled, volume and any observed impacts.	Adapted from GRI
T.C2.1	Impact driver	Pollution/pollution removal	Material recovery from wastewater	Weight (tonnes) of materials (e.g. minerals and metals) recovered from wastewater, and the proportion (%) this represents of the total quantity of those materials present in the wastewater before recovery.	TNFD

Metric no.	Metric category	Metric subcategory	Core sector disclosure indicator	Core sector disclosure metric	Source
T.C2.3	Impact driver	Pollution/pollution removal	Circularity of infrastructure	**Datacentres, Telecommunications infrastructure** Weight (tonnes) of network equipment or infrastructure installed during the reporting period that was reused or refurbished, and the proportion (%) this represents of the total weight installed during the reporting period.	Adapted from GSMA (2024)
T.C23.0	Response	Dependency, impact, risk and opportunity management: Changes to nature (dependency and impact): mitigation hierarchy steps	Reduction, reuse and recycling of water	**Semiconductors, EMS & ODM, Hardware and Datacentres** Volume (m^3) of water reused or recycled, and the reuse or recycling rate (%), calculated as water reused or recycled divided by the sum of water reused or recycled and new water intake. Water reused or recycled multiple times should be counted each time it is reused or recycled.	TNFD
T.C23.1	Response	Dependency, impact, risk and opportunity management: Changes to nature (dependency and impact): mitigation hierarchy steps	Wastewater reclamation	**Semiconductors, EMS & ODM, Hardware, and Datacentres** Quantity (m^3) and proportion (%) of wastewater reclaimed internally.	TNFD

Metric no.	Metric category	Metric subcategory	Core sector disclosure indicator	Core sector disclosure metric	Source
T.C23.2	Response	Dependency, impact, risk and opportunity management: Changes to nature (dependency and impact): mitigation hierarchy steps	Resource recovery and circularity	**Semiconductors, EMS & ODM, Hardware, Datacentres, Telecommunications Infrastructure** Weight (tonnes) and proportion (%) of high-impact commodities identified under C3.1 that are sourced as recycled/secondary materials or substituted with lower impact alternative materials, reported separately. Disclose the methodology and criteria used to determine lower environmental impact.	TNFD

3.3. Additional sector disclosure indicators and metrics

Table 12 includes additional disclosure metrics which are recommended for disclosure, where relevant, to best represent an organisation's material nature-related issues, based on their specific circumstances.

Table 12: Additional sector disclosure indicators and metrics

Metric no.	Metric category	Metric subcategory	Indicator	Additional sector disclosure metrics	Source
T.A1.0	Impact driver	Land/freshwater/ocean-use change	Land conversion	For organisations expanding their physical infrastructure: • Proportion (%) of new deployments using existing infrastructure and/or colocation rather than greenfield development.	TNFD
T.A1.1	Impact driver	Land/freshwater/ocean-use change	Restoration in the wider landscape/water basin	Number and area (km^2) of projects where an organisation has restored or contributed to the restoration of the wider water basin (e.g. upstream reforestation) beyond the land under direct operational control of the organisation.	TNFD
T.A2.2	Impact driver	Pollution/pollution removal	Packaging	**EMS & ODM, Hardware** Total weight of non-plastic packaging placed on the market (tonnes), and the weight and proportion (%) that is reusable and recyclable, respectively.	

Metric no.	Metric category	Metric subcategory	Indicator	Additional sector disclosure metrics	Source
T.A3.0	Impact driver	Resource use/ replenishment	Daily and monthly water withdrawal	**Semiconductors, EMS & ODM, Hardware and Datacentres** • Monthly water withdrawal (m^3); and • Peak daily water withdrawal (m^3), including calendar date.	TNFD
T.A3.1	Impact & response driver	Resource use/ replenishment	Cooling system operation	**Datacentre facilities using evaporative cooling systems** • Operating hours in each cooling mode: evaporative, mechanical/DX, chilled water and hybrid; • Water used per cooling mode (m^3); • Seasonal operation pattern (summer/winter cooling profile); • On-site water storage capacity (m^3); and • Maximum cooling duration without external water supply (hours).	TNFD

Metric no.	Metric category	Metric subcategory	Indicator	Additional sector disclosure metrics	Source
T.A3.2	Impact driver	Resource use/ replenishment	Metering coverage	**Datacentres** • Proportion (%) of water inflows metered rather than estimated; and • Proportion (%) of water outflows, including wastewater, that are metered. An organisation should provide details of sub-metering for cooling towers, adiabatic units, humidifiers.	TNFD
T.A3.3	Impact driver	Resource use/ replenishment	Water use of installed IT capacity	**Datacentres** Water use (m^3) per megawatt (MW) of installed IT capacity.	TNFD
T.A3.4	Impact driver	Resource use/ replenishment	Water Usage Impact (WUI)	**Datacentres** Water Usage Impact (WUI[103]) score, calculated using water consumed and the water stress level at the datacentre location (see Box 3).	The Green Grid (2025)

103 See Box 3 on Water efficiency and impact metrics for datacentres.

Metric no.	Metric category	Metric subcategory	Indicator	Additional sector disclosure metrics	Source
T.A3.5	Impact driver	Resource use/ replenishment	AI/HPC water intensity	**Datacentres** For Graphics Processing Units (GPU)-heavy sites: • Water use per GPU per hour (m^3/h); • Water use per kWh delivered to GPU racks (m^3/kWh); and • Water use per model training run (if the customer requests structured reporting) (m^3/run).	TNFD
T.A3.6	Impact driver	Resource use/ replenishment (multiple drivers of change associated with electricity generation)	Electricity consumption by source	**Datacentres** Total electricity consumption for datacentre operations (MWh) and proportion (%), disaggregated by source of supply, including: • Electricity imported from the grid; • Electricity generated on-site from renewable sources; • Electricity generated on-site from fossil-fuel sources; and • Electricity supplied through direct power purchase agreements or other dedicated off-site generation arrangements, where applicable.	TNFD

Metric no.	Metric category	Metric subcategory	Indicator	Additional sector disclosure metrics	Source
T.A19.0	Response	Strategy: Policies, commitments and targets	Science-based target for freshwater quantity/ landscape approach/ water withdrawal	Number and proportion (%) of sites in areas of water scarcity where an organisation has set a science-based target for freshwater quantity[104] and/or has worked with other stakeholders in the water basin to determine sustainable water withdrawal limits from the basin. Disclose relevant information on the stakeholder process, basin and withdrawal limits.	TNFD, SBTN
T.A23.0	Response	Dependency, impact, risk and opportunity management: Changes to nature (dependency and impact): mitigation hierarchy steps	Design for disassembly	**Semiconductors, EMS & ODM, Hardware, Datacentres, Telecommunications Infrastructure** Number and proportion (%) of new products, facilities, infrastructure and equipment developed or procured that are covered by defined design-for-disassembly, repair, refurbishment and component-recovery criteria, by product or asset category. Disclose the criteria or standard applied.	TNFD
T.A25.0	Response	Dependency, impact, risk and opportunity management: Dependency, impact, risk and opportunity assessment	Protected/sensitive natural areas screening	Proportion (%) of sites geolocated and screened for overlap with or proximity to protected and/or sensitive natural areas.	TNFD

104 SBTN (2024) Freshwater targets. Freshwater targets – Science Based Targets Network.

Metric no.	Metric category	Metric subcategory	Indicator	Additional sector disclosure metrics	Source
T.A25.1	Response	Dependency, impact, risk and opportunity management: Dependency, impact, risk and opportunity assessment	Environmental impact assessment coverage	**Semiconductors, Datacentres** Number and proportion (%) of new facilities and expansions subject to an environmental impact assessment that included assessment of impacts on biodiversity, ecosystems and habitat connectivity.	TNFD IFC Performance Standard 1: Assessment and Management of Environmental and Social Risks and Impacts IFC Performance Standard 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources

Annex 1: SICS® industries in scope

Box 4: SICS® industries in the scope of this guidance document

Semiconductors

Semiconductors industry entities design or manufacture semiconductor devices, integrated circuits, their raw materials and components, or capital equipment. Some entities in the industry provide outsourced manufacturing, assembly or other services for designers of semiconductor devices.

Electronic Manufacturing Services (EMS) & Original Design Manufacturing (ODM)

The EMS and ODM industry consists of two main segments. EMS entities provide assembly, logistics and after-market services for original equipment manufacturers. ODM entities provide engineering and design services for original equipment manufacturers and may own significant intellectual property. Although EMS and ODM entities produce equipment for a variety of sectors, the industry is associated closely with the hardware industry, which consists of entities that design technology hardware products such as personal computers, consumer electronics and storage devices for both personal consumers and businesses.

Hardware

Hardware industry entities design and sell technology hardware products, including computers, consumer electronics, communications equipment, storage devices, components and peripherals. Many entities in the industry rely heavily upon the EMS and ODM industry for manufacturing services. The industry is expected to continue to grow as technology use rapidly increases, especially among emerging market consumers.

Software & Information Technology (IT) Services

The Software & IT Services industry offers products and services globally to retail, business and government customers and includes entities that develop and sell applications software, infrastructure software and middleware. The industry generally is competitive but with dominant players in some segments. Although relatively immature, the industry is characterised by high-growth entities that place a heavy emphasis on innovation and depend on human and intellectual capital. The industry also includes IT services entities delivering specialised IT functions, such as consulting and outsourced services. New industry business models include cloud computing, software as a service, virtualisation, machine-to-machine communication, big data analysis and machine

learning. Additionally, brand value is important for entities in the industry to scale and achieve network effects, whereby wide adoption of a particular software product may result in self-perpetuating growth in sales.

Telecommunications Services

Telecommunication Services industry entities provide a range of services from wireless and wireline telecommunications to cable and satellite services. The wireless services segment provides direct communication through radio-based cellular networks and operates and maintains the associated switching and transmission facilities. The wireline segment provides local and long-distance voice communication via the Public Switched Telephone Network. Wireline carriers also offer voice over internet protocol (VoIP) telephone, television and broadband internet services over an expanding network of fibre optic cables. Cable providers distribute television programming from cable networks to subscribers. They typically also provide consumers with video services, highspeed internet service and VoIP. Traditionally, these services are bundled into packages that charge subscribers a single payment. Satellite entities distribute TV programming through broadcasting satellites orbiting the earth or through ground stations. Entities serve customers primarily in their domestic markets, although some entities operate in more than one country.

Internet Media & Services

The Internet Media & Services industry consists of two main segments. Entities in the Internet Media segment provide search engines and internet advertising channels, online gaming and online communities such as social networks. They also provide content, which is usually easily searchable, such as educational, medical, health, sports or news content. Entities in the internet-based services segment sell services, mainly through the internet. The industry generates revenue primarily from online advertising, usually on free content, with other revenue coming from subscription fees, content sales or the sale of user information to third parties.

From SASB Standards (2024) Sustainable Industry Classification System (SICS®)

Annex 2: ISIC groups/classes included in industry materiality ratings

Annex 2 provides an overview of the SASB industries and the associated ISIC division and group/class that have been mapped under each industry as part of the materiality ratings in Section 2.

Table 13: SASB industries mapped against ISIC division and group/class

SASB industry	ISIC Division	ISIC Group / Class
Semiconductors	Manufacture of computer, electronic and optical products	Manufacture of electronic components and boards
EMS & ODM	Manufacture of computer, electronic and optical products	Manufacture of computers and peripheral equipment Manufacture of electronic components and boards Manufacture of communication equipment Manufacture of consumer electronics
	Manufacture of electronic equipment	Manufacture of batteries and components Manufacture of wiring and wiring devices Manufacture of other electronic equipment
Hardware	Manufacture of computer, electronic and optical products	Manufacture of consumer electronics
	Wholesale and retail trade	Wholesale of machinery, equipment and supplies
	Retail trade	Retail sale of information and communications equipment in specialised stores

SASB industry	ISIC Division	ISIC Group / Class
Datacentres	Information services	Data processing, hosting and related activities
	Computer programming, consultancy and related activities	Computer programming, consultancy and related activities
Telecommunications infrastructure & services	Telecommunications	Wired telecommunication activities Wireless telecommunication activities Satellite telecommunication activities Other telecommunication activities
Software & IT services	Computer programming, consultancy and related activities	Computer programming, consultancy and related activities
	Information services	Data processing, hosting and related activities
	Wholesale and retail trade	Wholesale of machinery, equipment and supplies
Internet & Media services	Information service activities	Other service information activities Retail trade, not in stores, stalls or markets

Annex 3: List of pollutant classes and illustrative examples of substances of concern by sub-sector

This list provides sector-specific illustrative examples of substances of concern within each pollutant class and is not exhaustive. Organisations should refer to this list when disclosing against the relevant pollution metrics in Section 3. This list provides sector-specific examples and is not exhaustive.

Refer to Annex 4 for selected standards, regulations and frameworks relevant to pollutants. Where applicable legal or regulatory frameworks require disclosure of additional pollutants or impose more stringent measurement or reporting requirements, organisations should apply those requirements when reporting against the relevant TNFD metric.

Table 14: List of water-related pollutant classes and illustrative examples of substances of concern relevant by subsector

Pollutant class	Illustrative examples of substances of concern	Applicable sub-sector(s)
Per- and Polyfluoroalkyl Substances (PFAS)	PFOS and other PFAS compounds	Semiconductors, EMS & ODM, Hardware
Metals and Metalloids	Copper (Cu), Zinc (Zn), Nickel (Ni), Chromium (Cr), Lead (Pb), Cadmium (Cd), Mercury (Hg), Arsenic (As), Selenium (Se), Tin (Sn)	Semiconductors, EMS & ODM, Hardware, Datacentres
Organic Pollutants	Tetramethylammonium Hydroxide (TMAH), azoles, solvents, developers, surfactants, cleaning-agent residues, flux residues	Semiconductors, EMS & ODM, Hardware
Cyanides (CN^-)	Total cyanide, free cyanide	Semiconductors, EMS & ODM, Hardware
Nutrients	Ammonia (NH_3-N), nitrate (NO_3^-), Total Nitrogen (TN), Total Phosphorus (TP)	Semiconductors, EMS & ODM, Hardware
Halogenated organic compounds	Chlorinated and brominated organic compounds	Semiconductors
Fluoride Compounds	Fluoride	Semiconductors
Cooling-water Treatment Chemicals	Biocides, corrosion inhibitors	Datacentres

Annex 4: List of standards, regulations and frameworks relevant to pollution

Organisations should refer to selected standards, regulations and frameworks below when disclosing against the relevant pollution metrics in Section 3.

This list includes international conventions and regional and national frameworks relevant to pollutants, hazard classes and substances of concern, and is not exhaustive. Where applicable legal or regulatory frameworks require disclosure of additional pollutants or impose more stringent measurement or reporting requirements, organisations should apply those requirements when reporting against the relevant TNFD metric.

Table 15: Selected examples of environmental quality standards, regulations and frameworks for pollutants

List	Type of source	Summary
PFAS Toxic Release Inventory	Inventory	PFAS Toxic release inventory.
Restriction of Hazardous Substances in Electrical and Electronic Equipment (RoHS)	Regulation	RoHS restricts the use of ten substances: lead, cadmium, mercury, hexavalent chromium, polybrominated biphenyls (PBB) and polybrominated diphenyl ethers (PBDE), bis(2-ethylhexyl) phthalate (DEHP), butyl benzyl phthalate (BBP), dibutyl phthalate (DBP) and diisobutyl phthalate (DIBP).
EU REACH Regulation	Regulation	REACH lists 476 substances that, if exceeding 1 tonne per year per company, must be registered with ECHA. Companies must identify the risks linked to the substances they handle and indicate how they manage them. This obligation applies to both substances and mixtures.

List	Type of source	Summary
Appendix C of the EU Taxonomy Delegated Acts	Regulation	Sets the generic 'Do No Significant Harm' (DNSH) criteria for pollution prevention and control regarding the use and presence of chemicals. It restricts economic activities from manufacturing, placing on the market or using hazardous chemical groups like POPs, mercury and certain REACH-regulated substances.
Toxic Substances Control Act (TCSA)	Regulation	TSCA addresses the production, importation, use and disposal of specific chemicals including polychlorinated biphenyls (PCBs), asbestos, radon and lead-based paint. It contains more than 83,000 chemicals. Certain substances are generally excluded including, among others, food, drugs, cosmetics and pesticides.
Proposition 65 List	Regulation	Proposition 65 provides a list of naturally occurring and synthetic chemicals causing cancer and/or reproductive harm.
PFAS Global Portal for harmonised classification	Scientific and policy guidance	OECD's global portal provides a Harmonised Integrated Classification System for Human Health and Environmental Hazards of Chemical Substances and Mixtures.
Environmental health and safety (EHS) guidelines for semiconductors and other electronics manufacturing	Scientific and policy guidance	The EHS guidelines contain the performance levels and measures that are generally considered to be achievable in new facilities by existing technology at reasonable costs.
Chemicals Used in the Electronics Industry	Academic literature	Provides an overview of chemicals used in the electronics industry along with their hazards and toxicity concerns. It surveys regulatory and non-regulatory initiatives in place around the world that promote safer chemical selection and reutilisation of chemicals and materials and describes tools and practices that contribute to reduced consumption of chemicals of concern in this industry.

List	Type of source	Summary
Stockholm Convention on Persistent Organic Pollutants (POPs)	Regulation	List of POPs as listed in the Stockholm Convention for parties to take measures to eliminate production and use (Annex A), to restrict production and use (Annex B) and to reduce unintentional releases (Annex C).
EU Environmental Quality Standards (EQS) Directive (2008/105/EC), Annex I, Part A	Regulation	Part A of Annex I lists priority substances and sets environmental quality standards (EQS) for surface waters.
National Primary Drinking Water Regulation	Regulation	US EPA established legally enforceable levels, called Maximum Contaminant Levels (MCLs) for six PFAS in drinking water.
ISO 14046 Water Footprint	Technical standards	ISO 14046 specifies principles, requirements and guidelines related to water footprint assessment of products, processes and organisations based on life cycle assessment (LCA). Only air and soil emissions that impact water quality are included in the assessment, and not all air and soil emissions are included.
GRI 303: Water and effluents 2018	Technical standards	GRI 303 provides reporting requirements on water withdrawal, consumption, discharge and related impacts caused or contributed to, or that are directly linked to its operations, products or services by its business relationship (e.g. runoff).
The e-Stewards Standard	Technical standards	Certification standard for e-waste.
Seri's R2V3 Standards	Technical standards	Certification standard that ensures electronics recyclers and ITAD companies manage electronic waste responsibly, protecting the environment, ensuring data security and safeguarding worker health and safety.

List	Type of source	Summary
IEC 62474 Material declaration for products of and for the electrotechnical industry	Technical standards	Specifies the procedure, content and form relating to material declarations for products of companies operating in and supplying the electrotechnical industry. It provides data to downstream manufacturers that allows them to assess products against substance restriction compliance requirements that they can use in their environmentally conscious design process and across all product life cycle phases.
New Batteries Regulation	Regulation	EU regulation concerning batteries and waste batteries.
OECD Due Diligence Guidance for Responsible Sourcing of Minerals from Conflict-Affected and High-Risk Areas	Scientific and policy guidance	Provides step-by-step management recommendations endorsed by governments for global responsible supply chains of all minerals, in order for companies to respect human rights and avoid contributing to conflict through their mineral or metal purchasing decisions and practices.
EU Industrial Emissions Directive (2010/75/EU)	Regulation	EU regulation on industrial emissions (integrated pollution prevention and control).
OECD Emission scenario document for chemicals used in the electronics industry	Scientific and policy guidance	Provides information on the sources, use patterns and release pathways of chemicals used in the electronics industry to assist in the estimation of releases of chemicals into the environment.

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